T.C.
OSTIM TECHNICAL UNIVERSITY
Faculty of Engineering

Department of Electrical and Electronics Engineering

Design and Implementation of a Search and Rescue Robot Dog for Disaster Areas

(Graduation Project I)

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210202015 Sıla ÖZDEMİR
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Dr. Şenol GÜLGÖNÜL

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GRADUATION PROJECT APPROVAL FORM

The undergraduate graduation project titled "[Afet Bölgelerinde Arama Kurtarma Amaçlı Robot Köpek Tasarımı ve Uygulaması]", prepared by [Uygar BAŞ, Sıla ÖZDEMİR, Berna Meltem YILDIRIM, Ahmet Emirhan GÖKTÜRK] under the supervision of

[Dr. Şenol GÜLGÖNÜL] was examined and accepted as an Undergraduate Graduation Project in terms of its scope and quality.

Supervisor	:	Dr. Şenol GÜLGÖNÜL
Jury member 1	:	Dr. Burak YENİPINAR
Jury member 2	:	Prof. Dr. Sinan KIVRAK
Head of the Department	:	Prof. Dr. İsmail Hakkı ALTAŞ

FOREWORD

This graduation project was prepared by the students of the Department of Electrical and Electronics Engineering, Faculty of Engineering, OSTIM Technical University, within the scope of Graduation Project I. The project focuses on the design and implementation of a quadruped robot dog intended for search and rescue operations in disaster areas.

We would like to take this opportunity to express our thanks to Asst. Prof. Dr. Senol GULGONUL, our project supervisor and mentor, for the valuable support and guidance that he provided us during the development of our final project. His knowledge and encouragement were critical to successfully completing our graduation project.

We would also like to thank Professor Dr. Ismail Hakkı ALTAS, Head of the Electrical and Electronics Engineering Department, for providing us with an outstanding academic environment that enabled us to successfully complete our graduation project.

Finally, we would like to express our heartfelt appreciation to all our families for their continuous support, patience, and encouragement throughout our academic journey, including the completion of our final project.

December 2025

Uygar BAŞ
Sıla ÖZDEMİR
Berna Meltem YILDIRIM
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ABSTRACT

Search and rescue operations are greatly challenged by natural disasters, such as earthquakes, due to collapsed structures, unstable terrain, and limited access. The traditional rescuer, human, must take many risks and face time constraints when performing rescues in these types of environments. Robotic systems that can safely and effectively operate in abnormal, hazardous, and/or otherwise challenging environments are becoming more desirable in the field.

In this graduation project, the design and implementation of a quadruped search and rescue robot dog for disaster areas is presented. The proposed robot is designed to move effectively on uneven and debris-filled surfaces, where wheeled or tracked robots are insufficient. Owing to its four-legged structure, the robot is capable of maintaining balance and mobility in challenging terrains commonly encountered in disaster zones.

The prototype is composed of three functional categories: mechanical systems, electronics, and embedded systems; these are combined, through integration, to develop one prototype. The stability of the actuators (legs) enables the robot to maintain a consistent position and to walk in a smooth manner. In order to achieve a walking robotic device with balanced mobility, engineers used methods based on PID control. The prototype contains many different types of sensors, including an inertial measurement unit, a Lidar sensor, a camera, and a microphone, which will allow the robot to distinguish potential survivors from ordinary people. All of the sensor readings will be processed in real-time; therefore, except for occasional manual intervention, all tracking will occur automatically. Specific signal detection and locations will be sent wirelessly to a remote command and control facility.

In both cases of being semi-autonomous and fully autonomous, there will be a system with a level of flexibility based on each respective mission. The overall purpose of this project involves increasing the efficiency of search and rescue, improving safety for personnel, and still providing a means for increasing situational awareness in disaster areas. The developed prototype will be a low-cost, easily manufactured robotic solution that supports future research and development in disaster response robotic solutions.

SYMBOLS AND ABREVIATIONS

AI	Artificial Intelligence
CAD	Computer-Aided Design
DOF	Degree of Freedom
IK	Inverse Kinematics
IMU	Inertial Measurement Unit
I2C	Inter-Integrated Circuit
LiDAR	Light Detection and Ranging
Li-Po	Lithium-Polymer Battery
MCU	Microcontroller Unit
PID	Proportional–Integral–Derivative
PWM	Pulse Width Modulation
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver–Transmitter
W_total	Total weight of the robot (N)
W_leg	Load/weight per leg (N)
m	Mass (kg)
g	Gravitational acceleration (m/s^2)
F_leg	Force acting on one leg (N)
F	Applied force (N)
T, τ	Torque (N·m)
r	Lever arm / moment arm (m)
L	Distance from joint axis to load point (m)
σ	Normal stress (Pa)
M	Bending moment (N·m)
c	Distance from neutral axis (m)
I	Area moment of inertia (m^4)
A	Cross-sectional area (m^2)
x, y	Foot position coordinates (m)
l1, l2	Link lengths (e.g., femur and tibia) (m)

$\theta1, \theta2$ Joint angles (rad or deg)

m_i Mass of component i (kg)
x_i, y_i Position of component i (m)
x_cm, y_cm Center of mass coordinates (m)

P_total Total power consumption (W)
P_i Power of component i (W)
E Energy (Wh or J)
t Time / operating duration (h or s)

I_total Total current consumption (A)
I_servo Current per servo / servo group (A)
I_control Control electronics current (A)
C Battery capacity (Ah)

f_control Control loop frequency (Hz)
T_control Control loop period (s)

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1.INTRODUCTION

1.1. General Information

1.1.1. Global and Regional Impacts of Natural Disasters and the Need for Search and Rescue

Natural disasters have been part of human existence since there have been humans-causing great turmoil both socially and economically, as well as demographically.

While you cannot stop a natural disaster from occurring, you can lessen its effect. Increased frequency and destructiveness of natural disasters are a direct result of factors such as global warming, urban sprawl, increased population, and tectonic activity. The total cost to the world from natural disasters in the last 20 years has been in the trillions of dollars according to the United Nations Office for Disaster Risk Reduction (UNDRR).

In addition, many hundreds of millions of people have died or lost their homes due to natural disasters. Because of its geographical location on the Alpine-Himalayan seismic belt, Türkiye experiences the most seismic activity. The presence of three major fault lines- North Anatolian Fault, East Anatolian Fault, West Anatolian Fault contributes to over 90% of Türkiye's land surface being at risk for earthquakes. Recent Turkish history such as the 1999 Marmara earthquake, the 2011 Van earthquake and most recently, the Kahramanmaraş earthquakes of February 6, 2023 show the real vulnerability of the building stock of Türkiye and the importance of the post-disaster response process. Labeled the "disaster of the century," the Kahramanmaraş earthquakes have a much wider geographical range than most previous earthquakes along with an extensive level of devastation resulting from extreme winter weather that overwhelmed conventional search and rescue efforts.

One of the main principles in the disaster response process is the "Golden Hours" factor. The first seventy-two (72) hours immediately following a disaster is the time period that provides victims under rubble with the highest probability of survival.

Finding victims under rubble during this time frame minimizes life-threatening conditions such as hypothermia, dehydration, blood loss, and Crush syndrome. Unfortunately, chaotic collapse of reinforced concrete structures that create impassable rubble piles, minimize existing survival avenues (triangles of life), create unstable building components, and pose threats from aftershocks present barriers for search and rescue teams to access quickly and safely. Moreover, search and rescue are showing diminishing returns as intervention based solely on human effort deteriorates through physical fatigue, psychological trauma, and safety concerns, which eventually decreases the rescue effectiveness.

The next step is the use of robotic technology and automated systems. Search and Rescue Robots are employed to provide services such as mapping, exploration, audio/video communications, and transporting of medical supplies. They can be sent

into locations that are either unsafe for people to enter or too hazardous for them. These automated systems gather real-time, precise and useful information, thus limiting the uncertainty faced by the teams making decisions or conducting rescues following a disaster strike.

1.1.2. Evolution of Mobile Robotic Systems: Transition from Wheeled to Legged Architectures

When looking at how search and rescue robots have evolved, the first thing to notice is that "mobility" was the main focus for technology. Search and rescue robots were created mainly with wheeled and tracked designs.

Wheeled Robots: The main advantages of these vehicles are their high speed, energy efficiency, and controllability on flat, hard surfaces (e.g., asphalt, concrete, etc.). However, their mobility is impeded upon encountering objects with a size greater than that of the wheels or at the point of low-friction ground surfaces (e.g., mud, sand). Objects like iron bars, piles of debris, and steps/ledges that are common in debris fields represent major obstacles to wheeled robot navigation.

Tracked Robots: Robots equipped with tank-style locomotion have a much larger area of contact with the ground, thereby providing them with increased traction. Track-type robots have been developed to be utilized in military or search and rescue operations (e.g., the PackBot), and they have been shown to traverse stairs and navigate rougher terrain more effectively than wheel-type robots. Nevertheless, although track-type robots utilize the same principle of maintaining "continuous contact with the ground," they have a high chance of getting stuck in crevasses, on steep terrain, or on piles of debris with complex and irregular shapes (i.e., unstructured terrain). Furthermore, track-style robots are much heavier than their wheel counterparts, thus requiring much more energy to operate.

Investigating living organisms in their natural habitats indicates that the best approach to moving through challenging environments is to have "legs". Legged robots can contact the ground at a point, which allows them to use even the tiniest safe footfalls available in a given area as leverage to jump over obstacles that wheels and tracks cannot traverse, climb steep staircases, manoeuvre through narrow corridors, and maintain their balance while adjusting their posture on uneven terrain.

Among legged systems, the quadruped robot is the architecture that combines the highest level of balance with the highest level of speed. Quadruped robots have better static stability than bipedally (humanoid) robots (because they can keep three feet on the ground while the fourth foot is in the air). Moreover, quadruped robots can achieve a greater speed and provide a lower mechanical complexity compared to six or eight-legged (hexapod/octopod) robots.

For this reason, quadruped robots have become widely accepted as the platforms that are the best "adapted to terrain" and used in contemporary search-and-rescue literature.

1.1.3. Dynamics of Quadruped Robots and Control Theory

Quadruped robot design and control are much more difficult engineering problems than those for wheeled robots. A wheeled vehicle can be statically stable when the motors have been turned off; however, the legged robot requires constant (active) control to maintain balance while standing, walking or running. Therefore, the control issue has two main components: Kinematics and Dynamics.

Kinematic Modeling: Calculating the position of the robot's limb (leg) end-effectors in reference to the robot's center of gravity.

Finding the foot tip's position based on joint angles would involve using forward kinematics.

To calculate the joint motor angles that the foot tip should be at for it to reach a desired position (i.e., the next step point), we use inverse kinematics. In our application, we will have to perform inverse kinematics calculations thousands of times per second for every step of the robot in order to send angle commands to the motors.

Balance Control and PID: In order for the robot to stay upright as it moves around, we need to keep the Center of Mass (CoM) under control. When stationary, we need to keep the CoM within the area supported by its feet on the ground; however, as soon as the robot starts to make moves outside of the area supported by its feet (i.e., when the robot runs or jumps), other forces become important. Some of these forces are referred to as Inertial Forces, and some of the terms we use to describe them include Zero Moment Point (ZMP). To keep control over the balance of the robot, PID (Proportional-Integral-Derivative) Controllers are used for controlling balance. The PID algorithm uses accelerometer and gyroscope data from the robot IMU (Inertial Measurement Unit) to identify the robot's current roll and pitch angles.

P (Proportional): Apply a reverse force proportional to how much the robot has tilted.

I (Integral): If the robot constantly stands tilted to one side (steady-state error), correct this over time.

D (Derivative): When an impact occurs suddenly, assess the tilt's speed, and then react by dampening any oscillation that may have resulted. Tuning all three parameters correctly will guarantee the body of the robot will remain stable and parallel with the ground, even when traversing an uneven surface like a stone.

1.1.4. Perception, Mapping, and Artificial Intelligence

A modern state-of-the-art search-and-rescue robot needs to be an autonomous unit that perceives its environment and can independently make decisions and carry out those decisions. In instances where the communication infrastructure in a disaster zone has been destroyed, and there are large pieces of concrete preventing the use of radio waves, it is crucial for the robot to be able to continue its search-and-rescue mission without having to rely on a connection to the operator. Therefore, the primary elements that provide this ability to be autonomous are Perception and AI (Artificial Intelligence).

LIDAR (Light Detection and Ranging): The sensor technology utilizing Laser Based (Lidar) Sensor creates 3D Maps of your surroundings with Laser Pulses. The Lidar will identify the surrounding environment of the robot including walls, voids, and objects around the robot with a centimeter-level precision. The data gathered from the Lidar will then run SLAM (Simultaneous Localization and Mapping) Algorithms; which means that whilst the robot builds a map of an unfamiliar environment, it can also tell where on that built map it is located.

Image Processing and Deep Learning: Lidar can detect the shape of an object but cannot determine what it is. This is where a camera and AI will help us. Our project is using powerful embedded computers like the NVIDIA Jetson Nano to process real-time camera images. Deep learning models for object recognition (e.g. YOLO- You Only Look Once) have been trained using debris images to identify specific patterns such as a "human body", "hand", or "thermal signal of life".

Sensor Fusion: Sometimes data from just one sensor can give you an inaccurate picture (for example, dust can cause a camera to not be able to see) but by combining (i.e., fusing) all the data from four sensors (Lidar, Camera, IMU, and Microphone), the robot will get a better understanding of where and what is happening. If the robot hears a sound saying "help," it will turn toward it, confirm there is a clear path with Lidar, and visually confirm the victim with a camera.

1.1.5. Project Objective and Unique Design Approach

This thesis aims to design and create a prototype for a low-cost, non-industrial quadruped robot (a domestic robot dog) to serve the needs described above and assist in disaster scenarios. The prohibitive costs of existing industrial robot dogs (such as Boston Dynamics Spot) limit the general use of this technology. This project supports the philosophy of "Accessable Technology" and will produce the built mechanical components of the robot using 3D Printing (Additive Manufacturing) technology.

The unique design approach of the project is based on the following fundamental pillars:

1. **Modular and Repairable Mechanics:** The robot's body and leg parts are designed to be produced even in the field. A broken part can be printed and replaced within minutes.
2. **Hybrid Processor Architecture:** The Raspberry Pi/Microcontroller is used for low- level motor control, while the Jetson Nano is utilized for high-level artificial intelligence, computer vision, and image processing. Thus, the entire system operates more efficiently as the respective tasks are delegated to the

appropriate hardware platform based upon processing capabilities and available resources.

3. **Multi-Sensor Integration:** By using Lidar, Stereo Camera, and Sound Sensors together, the robot's "life detection" success under debris is maximized.

This project is comprehensive R&D and draws on multiple engineering disciplines including mechanical design, electronics hardware integration, embedded software and Artificial Intelligence to be able to contribute technologically to the enhancement of Disaster Response capabilities in Canada.

1.2.LITERATURE REVIEW

The Literature Review section analyzes the current state of national and international research regarding Search and Rescue Robots (SRR), Quadruped Walking Mechanisms (QWM), Balance Control Algorithms (BCA), and Sensor Fusion (SF). The analysis includes both chronological and thematic approaches to the existing SRRs, QWMs, BCAs, and SFs in order. The Literature Review establishes the scientific foundation for this project, highlighting how the proposed SRR is different from those previously researched.

1.2.1.Development of Quadruped Robots and Academic Studies

The earliest studies of quadrupedal robots can be traced back to the 1980's when Marc Raibert created the MIT Leg Lab, pioneering the development of robotic robots that would ultimately be an ancestor to modern day robotics. The first robots to walk dynamically were developed using single leg systems (aka hoppers). The way these robot's maintained their equilibrium was by constantly jumping. The studies conducted by Raibert provided the basis for the further development of advanced robotic technologies.

In their article titled "Development of Quadruped Walking Robots: A Review," Biswal et al. (2021) presented an extensive review of quadrupedal robots that have been developed over the last 20 years. Within their review, they provided a detailed comparison of various aspects of quadrupedal robot development, including kinematic structure (both mammalian and reptilian leg configuration), propulsion method (electric, hydraulic, pneumatic), and control method. Biswal found that electric motors are more appropriate for small to medium-sized quadrupeds regarding energy efficiency and precision of control than hydraulic motors; however, he noted that because the hydraulic motor has a significantly greater load-carrying capacity than the electric motor (i.e., the Boston Dynamics BigDog), the stronger option may be best for larger robotics, depending on the desired function. Because our project is for an exploration mission that will be performed for search and rescue, and because we're creating a robotic quadruped that will be portable, we are designing it with electric servo motors to be consistent with Biswal's recommendations.

MIT has developed "Mini Cheetah," which is viewed in the literature as an important milestone. The publication of Katz, Carlo, and Kim (2019) Mini Cheetah: A Platform for Pushing the Limits of Dynamic Quadruped Control (Katz et al., 2019) includes experimental tests showing dynamic movements (backflip, recovery after falling, running.) The greatest innovation of the Mini Cheetah is in the development of their own quasi-direct drive actuators for the legs. These motors provide a high torque density with a very low gear ratio, thus producing compliance. The Katz study has also shown that managing the legs of the robot should not only include position control, but also torque and force control. In our project we will be using standard servo motors, due to budget constraints, but will apply virtual spring-damper models in our software to try to provide a similar elastic response to that of our prototype.

1.2.2. Balance Control Algorithms and Gait Planning

To traverse rough terrains effectively, the balance controller of a quadrupedal robot requires a significant degree of balance and stability. To identify potential compensation techniques for the quadrupedal robot's balance after it has sustained an external force (e.g. over a stick), Sun et al. (2022), in their paper entitled, "Balance Control of a Quadruped Robot Based on Foot Fall Adjustment" examined how to adjust the footfalls of quadrupeds in reaction to a stimulus. The results were obtained using a combination of the Model Predictive Control (MPC) and conventional Proportional- Integral-Differential (PID) controllers. By being able to predict its future position, Sun's study allowed for the prediction of possible loss of balance and the identification of ways for the robot to take preventive measures. An additional key finding of this study was that to improve stability when walking on wet and/or inclined surfaces, the foot tip forces should be optimised. Due to the limitations on processing power related to our Jetson Nano/Raspberry Pi (as compared to the original study), rather than using MPC-style methodologies with high processing loads, we chose to use enhanced PID and Zero Moment Point (ZMP) approaches derived from Sun's work, but without the severe processing burden incurred by the use of the MPC approach.

Chen and Nguyen (2021), in "Adaptive Force-based Control for Legged Robots," describe a way of adapting a legged robot's control by using information about the stiffness of the ground to modify the robot's stiffness. The authors noted that when the robot encounters a soft surface (e.g., sand, snow), the robot should increase its leg stiffness (i.e., move more stiffly), while it should decrease its leg stiffness when walking on a hard surface (i.e., move less stiffly). This method improved the energy efficiency of the robot by 15%.

1.2.3. Sensor Fusion and Autonomous Navigation

Studies of the use of sensor fusion in the development of autonomous search and rescue robots indicate that there is evidence supporting the importance of sensor fusion for enabling robots to operate autonomously. Ko et al. (2022), in their article, "Lidar-IMU Sensor Fusion- Based SLAM for Enhancing the Autonomous Navigation Performance of a Search and Rescue Robot," indicated that in environments where the visibility of Lidar is compromised due to smoke or dust, the use of Lidar only would not provide adequate information, and therefore, IMU data (an Inertial Sensor) should accompany Lidar data. The results from their studies demonstrated that utilizing the Kalman Filter to combine the geometric map generated by Lidar with the orientation and acceleration information generated by the IMU resulted in a reduction of positional error to within centimeters of the actual position. The research discussed in this article formed the basis for the use of both Lidar and IMU in conjunction with each other as well as in combination with gmapping and hector_slam algorithms running through the Robot Operating System (ROS).

1.2.4. Low-Cost and Open-Source Robot Projects

Not only have high-tech Lab robots been documented in the literature, but there are also studies of affordable and basic Robots. According to the publication, 'The Low-Cost Open Source Semi-Autonomous Quadruped Robot' by García-Cárdenas et al. (2020), a Robotic Dog could be made entirely from 3D Printed components and Common Hobby Servos. This publication has provided one of the most fundamental reference points for our Project. Specifically, García-Cárdenas emphasised that while using low-cost servos makes it difficult to control their torque limits and gear backlash, this issue can be alleviated through proper Kinematic Design (Leverage Length Optimisation). Additionally, the Stanford Dogs and Pupper projects of Stanford University have shown how powerful Open-Source Hardware can be. Our Project is utilising an NVIDIA Jetson Nano along with Image Processing and Artificial Intelligence capabilities, which will convert the 'simple' Robotic vehicle (RC) into an autonomous Search and Rescue Assistant.

1.2.5. Life Detection via Sound and Image Processing

The primary purpose of search and rescue robots is to locate missing persons. According to Lamo and Bassy (2024), in "A Survey of Sound Source Localization and Detection Methods," sound is the most accurate source of information if there isn't a direct line of sight among the debris. They explored methods to determine Direction of Arrival (DoA) through a microphone array. Conversely, in their research "AI on the Road: NVIDIA Jetson-Nano Computer Vision", Popa et al. (2024) showed that Jetson Nano can effectively run advanced models for image interpretation (e.g., YOLO and SSD) in real-time using edge technology. Together, these papers provide the foundation for our "perception layer" of our project's overall system. Our project represents an innovative hybrid system that employs both sound processing and image processing methodologies in one transportable mobile device.

1.2.6.The Project's Place and Distinctiveness in Literature

As a result of the literature review, it was observed that;

1. Existing industrial robots (Spot, ANYmal) are very expensive,
2. Academic robots (Mini Cheetah) are generally aimed at laboratory environments,

3. Low-cost robots (Open Source projects) generally lack autonomy and artificial intelligence features.

The goal of the study outlined in this thesis is to fill in the gaps that exist in the literature associated with "Cost-Effective Hardware" and "Advanced Artificial Intelligence." Our project is distinguished from other similar projects described in the literature through the use of deep learning algorithms that process data collected from sensors integrated into a simple mechanical structure 3D-printed at a reasonable price point, using a combination of Lidar, stereo camera, and microphone, along with processing equipment designed for use with Jetson Nano.

1.3. NOVELTY

Literature reviews, as well as the results of various industry research, suggest that all current search-and-rescue robotic systems are divided into essentially two groups. At one end of the spectrum, there are expensive, high-tech industrial robotic systems (like those developed by Boston Dynamics, specifically their Spot platform) that are priced well beyond the reach of most members of supply chains (typically ranging from $70,000 to $150,000) and therefore are very rarely found or utilized in today's search-and-rescue operations. At the other extreme, there exists a second group of robotic systems consisting of very low-cost hobbyist- type systems that typically provide extremely limited processing power and very little autonomy.

The search-and-rescue robot for disaster zones that is described and developed within this thesis is intended to be a new, innovative engineering solution that fills the void between these two categories:

- The fundamental elements constituting the novelty of the project are detailed below:

Cost-Effective and Sustainable Manufacturing Model:

This project developed the Mechanical Body and Walking Parts utilizing exclusively the Additive Manufacturing (3D Printing) process, rather than the traditional methods of production seen in industrial robotics. This approach reduces the cost of this robot to approximately 5% of the costs associated with Industrial Robots; additionally, it provides "Repairable in the Field." The ability to manufacture and replace a leg limb that has been damaged in a disaster zone with a mobile 3D Printer on location offers the opportunity of sustainable operations without needing to establish a system of logistical supply chains. This feature provides assurance of the sustainability and ongoing operation of this project.

Hybrid Embedded System Architecture and Edge AI:

Numerous studies done by researchers use a single microcontroller for controlling robots and to offload heavy processing operations onto a remote server or computer (Cloud Computing) for efficient processing. When it becomes impossible to communicate with a server remotely due to damage or other issues in a disaster area, the possibility of processing data using remote servers becomes unreliable. As a solution to this challenge, a hybrid framework was developed using both NVIDIA Jetson Nano (High Level) and Raspberry Pi or an Alternative Microcontroller (Low Level) to create the best of both worlds. The NVIDIA Jetson Nano will perform image recognition locally via Edge AI without any Internet connection, and the Low Level Processor (Microcontroller Unit) will provide control of motors and provide balancing functionality in the range of milliseconds. Distributing these tasks between two processors provides the most efficient and immediate response to any situation encountered by the robot.

Multi-Layer Sensor Fusion for Life Detection:

Most current robotic systems use only one type of data to operate. In this project, we present a multi-layered approach that combines visual (camera), spatial (Lidar) and auditory (microphone array) sensing. For example, in situations when the robot does not have visual access to its surroundings due to debris blocking its view, the robot's ability to locate the position of a call for help (sound source localization) using microphones and then focus the camera and Lidar in that direction reduces the likelihood of false positives related to identifying living victims.

Enhanced Control Algorithm for Dynamic Terrain Adaptability:

The non-precision torque fluctuation and gear backlash of low-cost servos present challenges for gait control. An adaptive PID Controller was created that works in conjunction with an inertial measurement unit (IMU) to provide software compensation for these hardware challenges. The adaptive PID controller prevents the robot from tipping during a foot slip or collapse of the ground, as the velocity of the centre of mass can be transferred to the remaining leg within milliseconds.

1.4. METHOD

A systems engineering methodology has been utilized to manage and carry out this project as defined by the systems engineering V-Model. The V-Model is a hierarchical system engineering process used to document/ define a project through the various phases of requirements definition, design and implementation, testing, and verification. The project will consist of four major project phases as follows:

Phase 1: Mechanical Design and Kinematic Analysis,

Modeling: SolidWorks 3D CAD software was used to create the body and limb components of the robot. In developing the design of the robot, we incorporated the Center of Mass (CoM), which has to be located low to avoid tipping over when walking over obstacles/debris, and also included an increase in the Range of Motion (RoM) for the robot's legs to allow for the ability to climb stairs.

Analysis: Using ANSYS's Finite Element Analysis (FEA), the durability of the parts was determined using stress concentra-tions that were assessed at the leg joints. The results of the analysis were used to optimize the infill density for 3D printed parts.

Kinematics: The inverse kinematics model for the robot was developed using a series of mathematical calculations to obtain a matrix equation that would calculate how many degrees each joint motor needed to rotate in order for the robot's body to reach the desired position (x, y, z).

Phase 2: Electronic Hardware Integration

Power System: The robot will be powered through the use of several lithium polymer (Li- Po) batteries which provide very high current capabilities. The mechanical drive train/battery pack is powered by a low voltage rating whereas the more sensitive electronic components will be powered through voltage regulation devices (buck regulators).

Communication: Sensors use I2C, microcontrollers communicate with UART, and cameras and lidar operate through USB protocols. Telecommunications data will also be sent from the robotics to the ground station using the 4G Module via Wireless Communication and WiFi.

Phase 3: Software and Algorithm Development

Robot Operating System (ROS): Modularity is the main component of the software framework being built using the Robot Operating System (ROS). All sensors and actuators will operate as ROS nodes and communicate with one another.

Image Processing: Image processing methods are created using Python programming language and OpenCV software library. Pre-trained YOLOv5 and/or MobileNet SSD models for detecting humans will be optimized with TensorRT and executed via Jetson Nano.

Balance Control: Euler angles (Roll, Pitch, Yaw) obtained from an IMU sensor will be used to generate real-time control signals for the motors via a PID feedback loop.

Phase 4: Test and Verification

Simulation: Prior to the robot being built physically, the design of the robot will be simulated within a Gazebo environment to allow for gait algorithms to be evaluated in a controlled setting.

Field Tests: Once the first prototype of the robot is manufactured, it will be subjected to a simulated debris test using a synthetic training track consisting of inclined ramps, rocky surfaces, and stairs, in a laboratory setting. Testing will measure key performance metrics (distance traveled before tipping over, battery run time, image latency, and detection accuracy rate) for the respective testing parameters.

1.5. IMPACT

Upon successful implementation of the project, the anticipated technical, economic, and social deliverables will be as follows:

1.5.1. Social and National Security Impact

Türkiye is situated in an earthquake-prone region, thereby making search-and-rescue technology a necessity. The robotic technology developed as part of this research project will enable search-and-rescue operations to be conducted prior to human personnel entering hazardous areas, damaged buildings, and/or tunnels that may be leaking gas (to conduct "vanguard reconnaissance"). By advancing the search-and-rescue operation prior to the arrival of human personnel, it is expected that due to the earlier identification of victims prior to entering the area, (1) the safety of search-and-rescue personnel will be enhanced; (2) the likelihood of locating victims trapped beneath debris will improve based on the ability to intervene in the first 96 hours after an earthquake; and (3) an increase in operational efficiency will occur by eliminating wasted response time in searching for victims in "hot zones."

1.5.2. Economic and Commercial Impact

Currently, search-and-rescue robots (produced by foreign manufacturers) used by AFAD (the Turkish Emergency Services) and/or Fire Brigade units/Authorities consume a substantial amount of foreign currency; this project will enable Türkiye to manufacture its search-and-rescue robots using local resources at significantly lower costs, thereby improving Türkiye's economy through the reduction of reliance on foreign products. The prototype developed through this project has excellent commercial potential and may be used in a variety of civilian and non-civilian applications, including the Defense Industry, the Mining Sector, and/or the Industrial Inspection sector.

1.5.3. Academic and Scientific Impact

The information gained from the project (i.e., performance of sensor fusion and strength tests on 3D printed parts) will add to the body of knowledge within the academic area of Mechatronics and Robotics. The findings will be reported in the form of papers presented at both domestic and foreign conferences and will also provide an example of "applied robotics" for other students attending the university. In addition, the Embedded System, Artificial Intelligence, and Mechanical Design skills gained through this project will assist the country in meeting its demands for qualified engineers.

1.6. STANDARDS

The International/Electricity standards provide an overview's Reliability, Safety and Conformity through Design and Build of a Robotic Society.

ISO 13482 (Robots and robotic devices — Safety requirements for personal care robots): The design of a robot must include Safety Procedures (Collision Avoidance) to reduce loss of life while the robot responding to requirements to Evacuate Victims safely.

IEC 62133 (Secondary cells and batteries containing alkaline or other non-acid electrolytes): Battery Management Systems will ensure that Lithium Ion Batteries are protected during Charge/Discharge cycles and against Overheating and Short Circuits.

IEEE 802.11 (Wireless LAN Standards): A robot must have Wireless (data & image) communication with Ground Control Stations in accordance with the Wireless Radio Frequency (2.4Ghz, 5Ghz) specifications.

IPC-2221 (Generic Standard on Printed Board Design): Electronic Circuit Board Design, Power Distribution Board Design, etc., is done according to IPC or other applicable IC Standards for Electrical/Signal Integrity.

ROS REP (Robot Operating System Enhancement Proposals): During Software Development, the ROS Community Naming/Package Structures (REP-103, REP-105) will be used for Code Reusability and Modularity.

Ethical Standards:

The principles of personal data privacy (GDPR/KVKK) and ethical guidelines have been upheld while using image processing technology during the course of this project. In addition, it has been accepted that any images collected will only be processed for use in search and rescue operations.

1.7. Work Schedule

The path to be followed in the Engineering Design and Graduation Project process is given in **Table 1.1** as a work-time graph.

In Table 1.1, the project is divided into work packages, and the work done in each package and the time intervals in which these series must be completed are specified. In order to avoid disruption of the work, the time period that must be done is given in the work package. In case any problems that may arise, plans B were considered and the risk analysis in **Table 1.2** was carried out.

Table 1.1. Work-Time Table

WP No	Name and Purpose of the Work Package	Students Responsible	Time Range	Contribution to the Project
1	Completion of the literature review and determination of unique design objectives for the project.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01/10 /2025-12 /11 /2025	Conducting a literature review, analyzing similar search and rescue robots, and determining system requirements.
2	Completed 3D model of the chassis and leg structure.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /10 /2025-01 /12 /2025	Modeling the mechanical design in a CAD environment, and performing center of gravity and stability analyses.
3	Completion of circuit schematics and component selection.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /12 /2025-01 /02 /2026	Creating the integration plan for electronic components (motor driver, sensors, microcontroller, 4G modem, GPS) and drawing circuit schematics.
4	Completion of the mechanical prototype.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /12 /2025-01 /02 /2026	Manufacturing mechanical parts via 3D printing and assembling the robot chassis.
5	Determination of PID parameters and observation of stable motion in simulations.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /01 /2026-01 /04 /2026	Developing the PID-based balance control algorithm and conducting simulation tests.
6	Achieving simultaneous processing of sensor data.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /03 /2026-01 /05 /2026	Integration of sensor systems (IMU, LiDAR, camera, microphone) and performing image processing tests on the Jetson Nano.
7	Establishment of the robot's basic locomotion (movement) capabilities.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /02 /2026-01 /05 /2026	Setup of the embedded system, software integration, and conducting locomotion (walking) tests.
8	Ensuring low-latency (lag-free) data transmission.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /03 /2026-01 /05 /2026	Data transmission tests via 4G modem and GPS, and optimization of wireless communication.

Table 1.1. Work-Time Table (Continuing)

WP No	Name and Purpose of the Work Package	Responsible Students	Time Range	Contribution to the Project
9	Stable operation of the robot in semi-autonomous and autonomous modes.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /04 /2026-01 /05 /2026	System testing: stability, sound detection, environmental mapping, and verification of autonomous movement.
10	Completion of the final report and prototype presentation.	Berna Meltem YILDIRIM Sıla ÖZDEMİR Uygar BAŞ Ahmet Emirhan GÖKTÜRK	01 /04 /2026-31 /05 /2026	Preparing the final report, evaluating project outputs, and preparing for the presentation.

1.8. Risc Analysis

Table 1.2. Risc analysis and plan B

WP No	Contribution to the Project	Risc Analysis
1	Instability in walking (locomotion) due to weight imbalance in the mechanical design.	PID control parameters will be retuned; mass distribution will be optimized through simulations, and the leg design will be updated. If necessary, the balance sensor (IMU) calibration will be repeated.
2	Delays in the procurement of electronic components (motor drivers, sensors, 4G modems, etc.).	Components with identical technical specifications will be sourced from alternative local or online suppliers. Equivalent parts will be used for temporary testing when necessary.
3	Occurrence of data synchronization problems between the Jetson Nano and Raspberry Pi during software integration.	Data transmission speed will be reduced to lower the processing load, and communication protocols (UART/I2C) will be reconfigured. If necessary, a data buffer will be added to prevent synchronization latency.

Table 1.3. Risc analysis and plan B (continuing)

WP No	Contribution to the Project	Risc Analysis
4	Interruption of the 4G modem connection or signal attenuation in a disaster environment.	In case of connection loss, the robot will temporarily store data in local memory and transmit it once the connection is re-established. Alternatively, short-range backup communication will be provided via Wi-Fi or LoRa modules.
5	Power insufficiency or battery failure during prototype testing.	A lithium battery protection circuit will be used, and operations will be supported by an external power supply during the testing process. The battery level will be continuously monitored by the software.

1.9. Organization of Work packages

For the project to progress in a planned and timely manner, a different person was designated as the leader in each work package respectively. Each student carried out the work package with discipline. Table 1.3 shows the work packages that each student led and the number of work packages they were involved in.

Table 1.3 Distribution of Work Package

Work Package Leader	Leding Work Package	Number of Work Packages
Sıla ÖZDEMİR	1,5,9	10
Uygar BAŞ	2,6,10	10
Ahmet Emirhan GÖKTÜRK	3,7	10
Berna Meltem YILDIRIM	4,8	10

2. THEORETICAL BACKGROUND

2.1. General Information

As designing our quadruped robot prototype, we knew that its main goal was to navigate complex terrain. A Jetson Nano, Raspberry Pi 5, an Inertial Measurement Unit (IMU), DepthAI OAK camera, and a battery form the backbone of this robot. For the robot's skeleton, ABS filament was used to ensure a lightweight structure while maintaining sufficient mechanical strength, allowing the robot to perform demanding physical tasks without sacrificing agility. Coming heading through tight spaces, rough terrain and uneven surfaces are no problem for the robot thanks to its carefully developed balance and movement algorithms.

The control interface that we're building is basically user-friendly and puts the focus on the person operating the robot, providing a crystal-clear view of the robot's condition and the environment it's operating in. We're also working on a real-time planning and monitoring system that lets operators monitor the progress of the robot and stay one step ahead of the game.

The robot's movement is achieved through coordinated control of its leg and shoulder joints, while environmental interaction and balance are maintained using sensor feedback.

2.2. Quadruped Robots and Robot Dynamics

Quadruped robots are characterized by their ability to maintain stability and mobility through coordinated leg movements. The motion of the robot is defined by the equations of motion governing the relationship between joint torques, link masses, and resulting accelerations. In this study, simplified dynamic models are considered to ensure stable and computationally efficient locomotion.

2.3. Kinematic Modeling of Robot Legs

Each robot leg is modeled as a serial kinematic chain. Forward kinematics is used to calculate the foot position based on joint angles, while inverse kinematics determines the required joint angles for a desired foot trajectory. Only the final kinematic relationships used in the control algorithm are presented in this section.

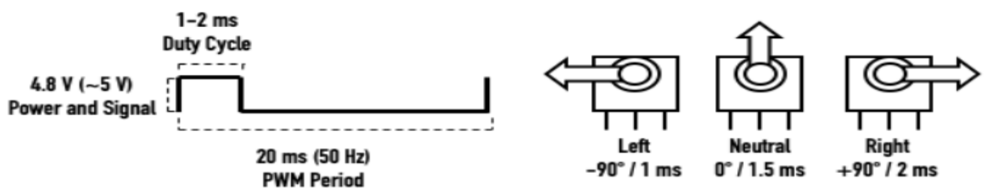


Figure 2.3. Calculation.

2.4. Robot Drive and Motors (Servo Motors)

The physical implementation of the robot's motion control system was executed using a Raspberry Pi 5 as the primary processing unit, interfaced with a PCA9685 16-channel 12-bit PWM driver. The robot is actuated using high-torque metal-gear servo motors, which were specifically selected for their mechanical durability and high positional accuracy under load. The operation of these actuators is governed by Pulse-Width Modulation (PWM), where a mathematical relationship between the signal's duty cycle and the joint's angular position serves as the foundation for the motion control algorithms.

As illustrated in the connection diagram provided in this section, the hardware architecture utilizes the I2C communication protocol via the SDA and SCL pins of the Raspberry Pi 5. The integration of the PCA9685 driver is a critical design choice, as it offloads the PWM generation from the main CPU, providing a high resolution of 4096 steps for precise positioning.

Furthermore, to ensure system stability and comply with electrical safety standards, the power distribution for the servo motors is separated from the logic power of the microprocessor. An external power source is connected directly to the PCA9685 terminal blocks; this configuration prevents voltage drops and protects the Raspberry Pi from potential electrical noise or overcurrent generated by the motors during high-torque maneuvers. This integrated approach ensures a robust interface between the software-defined motion commands and the physical movement of the robot joints.

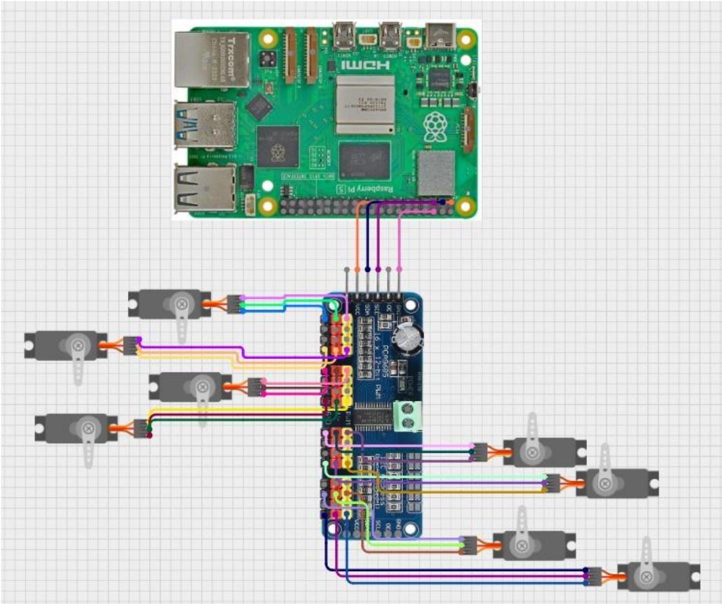


Figure 2.4. Servo Motors and Raspberry Pi 5.

2.5. Sensors and Jetson Nano's Feedback Systems

Data integrity and low latency are ensured by the integration of sensors according to specific industrial communication standards.

I2C Protocol (IMU Interfacing): The IMU - MPU-9250/6050 - is interfaced through the I2C protocol, using SDA/SCL lines. This protocol enables the processor to read the 9-axis motion data of an accelerometer, gyroscope, and magnetometer with a shared two-wire bus.

UART/Serial Communication (Lidar and GPS): The Lidar sensor provides high-frequency data, while the GPS module provides information used for positioning, both sent over Universal Asynchronous Receiver-Transmitter protocols. In the system diagram, converters from USB to TTL were used as a means of bridging these sensors with the USB ports on the Jetson, ensuring that serial data can be stably transmitted at high baud rates.

Camera Interface: MIPI CSI The Raspberry Pi Camera Module interfaces through the Camera Serial Interface (CSI), providing a direct link to the Image Signal Processor (ISP) within the Jetson module, thereby minimizing CPU overhead during high-definition video capture.

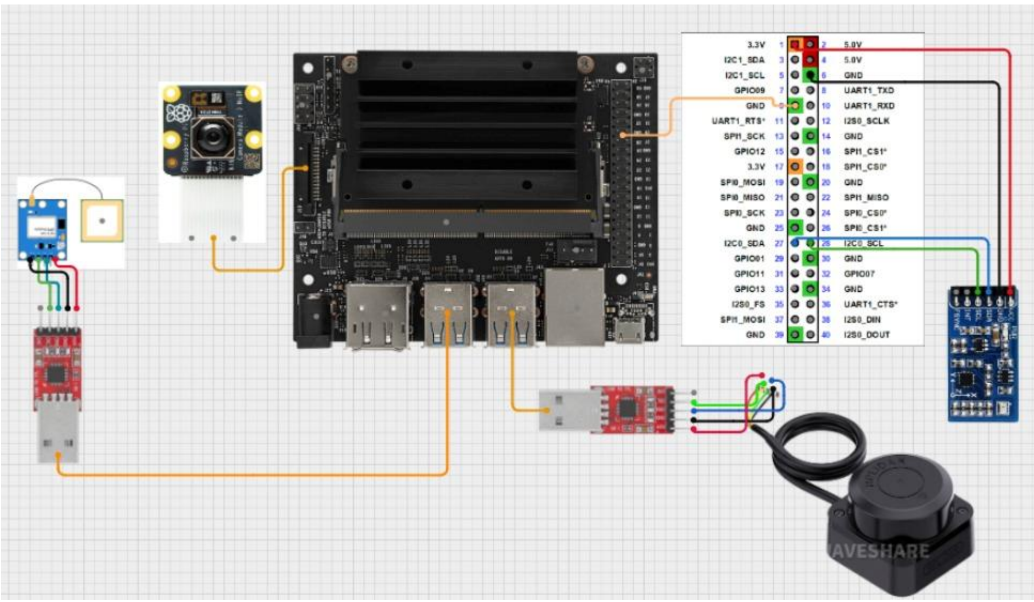


Figure 2.5. Jetson Nano and Sensors

2.6. Theory of Perception and Localization

Autonomous movement involves the simultaneous processing of both local and global coordinates.

Lidar Theory: The Lidar sensor makes use of the time-of-flight principles, emitting laser pulses and measuring the reflection time to estimate distances, thus returning a 2D point cloud of data for obstacle avoidance.

GPS Localization: The GPS module employs trilateration of satellite signals, yielding global latitude and longitude coordinates, which are necessary for long-range navigation.

2.7. Wired–Wireless Communication System

Wired and wireless communication methods are used in the proposed quadrupedal robot system to ensure reliable data exchange between the processing units, peripheral devices, and user interface. The communication architecture segregates high-bandwidth data transmission from low-level control communication for better stability and performance of the overall system.

Wired communication allows transmitting data over short distances with high reliability between the onboard components. As can be seen in the system connection diagram, Jetson Nano communicates with peripheral modules through USB and serial interfaces.

The Jetson Nano uses a USB interface to connect to the external communication module. The connection allows for two-way traffic of data between modules using serial communication principles. Serial communication permits structured packets of data to be transmitted sequentially, ensuring low latency and deterministic behavior for data related to control.

The general model for serial transmission of data may be represented as:

Encoder → Serial Channel → Decoder → Data → Rx

Where transmitted data is encoded into serial frames and reconstructed on the receiving side.

Wired connections are used within the robot due to their immunity against electromagnetic interference and predictable timing characteristics.

2.8. Wireless Data Communications

Remote monitoring and multi-machine communication are enabled by high-speed wireless protocols. The system uses a TP-Link Archer T4U dual-band USB adapter.

Communication Standard**:** The module utilizes the IEEE 802.11ac (Wi-Fi 5) standard, necessary to transmit high-bandwidth data such as real-time video feeds and Lidar point clouds.

Dual-Band**:** Operating on both 2.4 GHz and 5 GHz frequencies, the system minimizes interference. The 5 GHz band is exclusively used for low-latency control commands and high-speed data throughput.

Network Integration**:** In the system diagram, the Jetson is connected with the local network using a router, which enables wireless teleoperation and remote debugging from a secondary workstation.

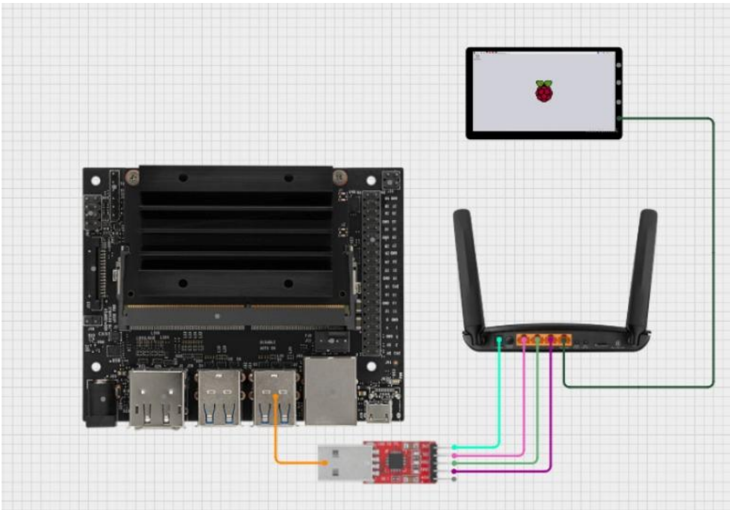


Figure 2.8.Wi-fi Modul and Jetson Nano

2.9. Microprocessors and Distributed Computing Architecture

The multi-processor approach is used in system architecture to balance highly computationally intensive tasks with peripheral management. The main processing hub is an NVIDIA Jetson module interfaced with a Raspberry Pi 5 for creating a distributed computing environment.

NVIDIA Jetson (Master Unit): The module provides GPU-accelerated computing; this is employed for heavy tasks of real-time Lidar data processing, image recognition via the Raspberry Pi Camera Module 3 NoIR, and autonomous path planning.

The Raspberry Pi 5 is an integrated secondary unit that aids in general purpose computing, including the ability to add more I/O flexibility in sensor management.

Inter-Processor Communication (UART): The system wiring diagrams show that Jetson and Raspberry Pi 5 are in sync through a UART link. This connection uses Pins 8 and 10 for TXD and RXD, respectively, and Pin 39 for GND in the header of Jetson to establish serial data communication between both platforms for low-latency command transmission.

2.10. Control Methods

A hierarchical control architecture was implemented to manage the complex movements of the quadruped robot while maintaining balance. The core of this system is based on Inverse Kinematics calculations. In this model, it calculates the required joint angles or each leg to move the robot's body to a specific coordinate or execute a step.

Joint Control and Precision: The calculated values of angles are sent across the I2C bus to the PCA9685 PWM driver. The driver transforms the data into PWM signals with a resolution of 12-bit, which is processed by the servo motors in order to achieve the desired positions.

Dynamic Stability: The acceleration and gyroscope data from the IMU sensor are processed continuously during locomotion to prevent the robot from tipping over. If the tilt about the pitch/roll axes of the robot exceeds its preset limits, the real-time update in leg lengths by the control algorithm aims to keep the center of mass within the support polygon.

2.11. Human-Machine Interaction

The module will allow for operational flexibility through remote monitoring and command transmissions via a wireless interface. This interaction layer required high-bandwidth hardware to be selected in order to provide continuous and smooth data exchange.

Wireless Communication Infrastructure: The data transmission on the robot is supported by the TP-Link Archer T4U Wi-Fi adapter, according to the IEEE 802.11ac standard. Choosing the 5 GHz band enables low-latency control commands to get through without the signal interference common in the 2.4 GHz spectrum.

Telemetry and Visual Feedback: The operator will have the ability, on a remote terminal, to view in real-time video from the Raspberry Pi Camera Module 3 NoIR and live occupancy grid maps derived from Lidar. This is so the operator can visually observe the environmental perception of the robot and take any necessary intervention in its navigation process.

2.12. Intelligent Systems

The "intelligence" of the robot is defined by its ability to perceive the environment around it and make autonomous decisions. This capability is realized through the high computing power of the NVIDIA Jetson platform combined with sensor fusion techniques.

SLAM and Autonomous Navigation: The robot performs a SLAM by fusing distance data from the RPLidar sensor with the orientation data from IMU. In doing so, the robot builds a map of an unknown environment while at the same time calculating its own coordinates in that map.

Obstacle Detection and Path Planning: The GPU cores in the Jetson module process the Lidar point cloud to identify obstacles that lie in the path of the robot. If an obstacle is found, the system automatically recalculates the trajectory toward the target GPS coordinates by the safest possible path. This level of autonomy enables the robot to perform missions in complex terrains without continuous human intervention.

2.13. Power Management and Energy Distribution System

The reliability of a quadruped robot depends largely on its power management system, since it needs to provide stable voltage to sensitive electronics while supplying high current for multiple servo motors during locomotion.

Energy Source (4S LiPo Battery): A 14.8V 4S LiPo battery with an 8000mAh capacity was chosen as the prime source of energy. In mobile robotics, LiPo batteries are preferred due to their high energy-to-weight ratio and the high discharge rates needed for the sudden torque requirements of the robot's legs. The capacity of 8000mAh will ensure extended mission times of the system dealing with the combined power draw contributed by Jetson, Raspberry Pi 5, and the actuation layer.

Voltage Regulation (Buck Converter): Since the logic units (Jetson and RPi 5) and the servo motors operate at lower voltage levels (typically 5V to 7.4V) than the battery's nominal 14.8V, a high-efficiency Buck Converter (Step-Down Regulator) is utilized.

High-Current Capabilities: The converter is specified for an 8A output to accommodate the peak current consumption. In a quadruped system with 8 or more high-torque metal-gear servos, the simultaneous movement of multiple joints can create significant current spikes. The capacity of 8A ensures that the voltage remains stable, without dropping, and thus prevents "brownouts" or unexpected resets of the microprocessors.

Safety and Efficiency: The architecture of power distribution is such that heat dissipation is minimized and electromagnetic interference is minimal. The stepping down of voltages closer to the load ensures high efficiency, hence making sure that maximum energy is directed toward the movement and processing tasks within the robot.

3. DESIGN

3.1. General Information

This section explains the engineering design of the proposed quadruped search and rescue robot developed for operation in disaster areas. The design process is based on mechanical, electrical, and software engineering principles, and aims to create a robust, mobile, and energy-efficient robotic platform capable of navigating uneven terrain such as rubble, debris, and collapsed structures.

A quadruped configuration was selected because of its superior performance (greater stability than wheeled or tracked systems) and adaptability to different surfaces. A quadrupedal robot design draws on existing quadrupedal robot systems as well as an existing open-source reference, but the overall geometry has been modified to fit the requirements of a disaster relief scenario; the dimensions (based on existing designs), materials (based on current technology), actuator placements (based on user specifications), and system integration (based on user requirements) have all been redesigned.

During the design phase, key engineering calculations are carried out to support critical decisions such as:

- Number and placement of servo motors
- Load distribution on each leg
- Required torque for joint actuation
- Power consumption and battery capacity
- Structural strength of 3D-printed components
- Control and communication architecture

Engineering of mechanical components is conducted through the use of

Computer Aided Design (CAD) software. Technical drawings will be created; these will include the mechanical component with full numerical dimensioning, as well as the project name, preparation date, designer names, and academic supervision. Electrical design and software design will be created in conjunction with the mechanical design to ensure compatibility of the entire system.

A comprehensive compilation of component parts is compiled at the conclusion of the design phase, along with an initial investigation into costs. Additionally, the legal and regulatory implications regarding the implementation of robotic systems in areas of public interest and disaster response are also conducted.

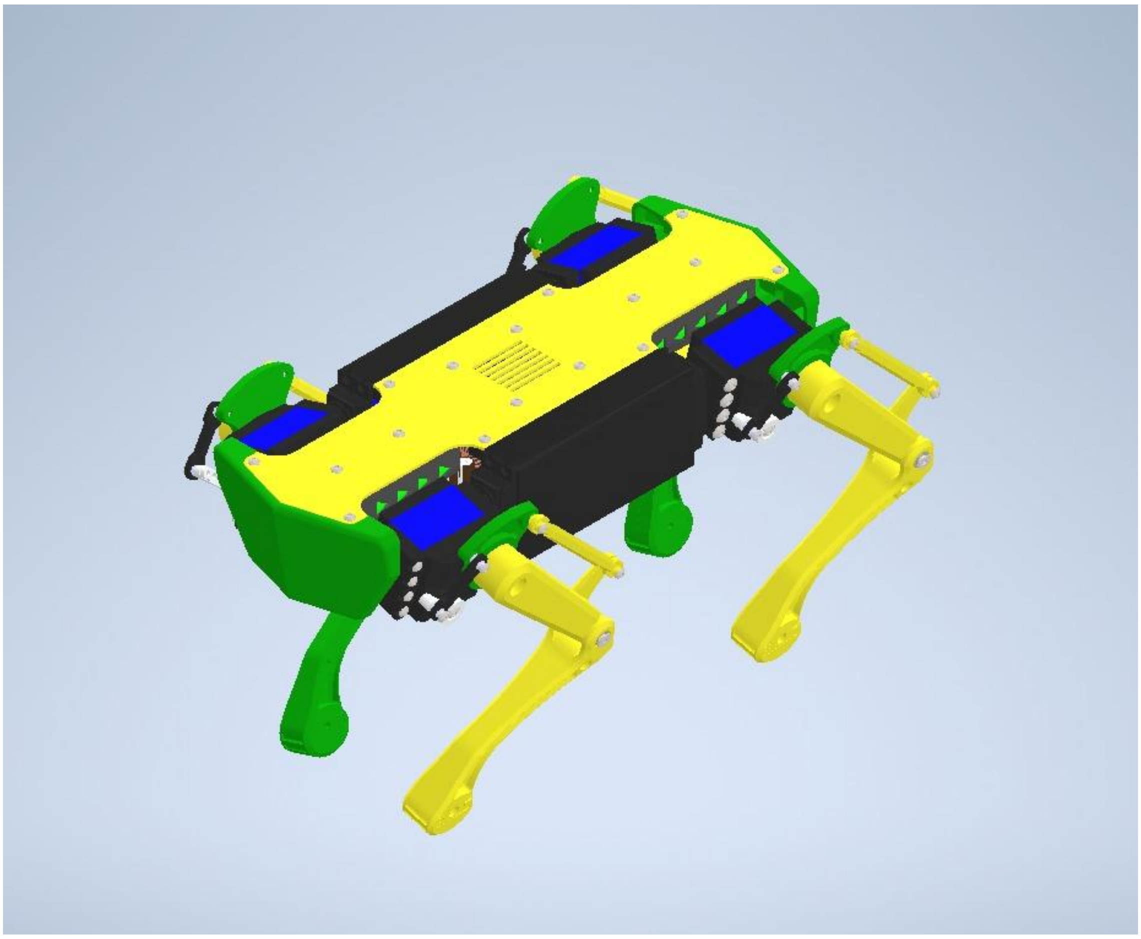


Figure 3.1. Overall CAD model of the quadruped robot

3.2. Dimensioning

The physical dimensions of the quadrupedal robots are established in accordance with their ability to successfully navigate restricted or cluttered environments (e.g., earthquake debris) while providing an adequate shield for the electronic components that lie within them. The chassis of each robot is configured as a compact unit, and the control unit, power electronic components, and battery arrangement are integrated so that they will have a location to rest on the robot's body, and, at the same time, will be designed with a low centre of gravity to improve the robot's (overall) stability.

The approximate body dimensions of the robot are planned as follows:

- **Body Length:** approximately **300 mm**
- **Body Width:** approximately **120 mm**

These dimensions allow adequate internal space for electronic components without significantly increasing the overall mass of the system.

Each leg is designed as a multi-segment structure to enable flexible and stable locomotion. The approximate lengths of the leg segments are:

- **Coxa:** approximately **40 mm**
- **Femur:** approximately **110 mm**
- **Tibia:** approximately **120 mm**

The robot is anticipated to have an overall height from ground to top of robot between approximately 15 cm to 20 cm. Depending on the robots posture and gait, this height may vary somewhat. In total, the robots weight- which includes the mechanical structure, electronics, battery, and sensors- is projected to be roughly 3,5 kg.

The dimensions listed in this section are illustrative only and serve as a reference for the design of the robot and the development of the associated technical drawings. As designs progress through a more detailed CAD model and integration of components further refinements will be made to achieve final dimensional specifications.

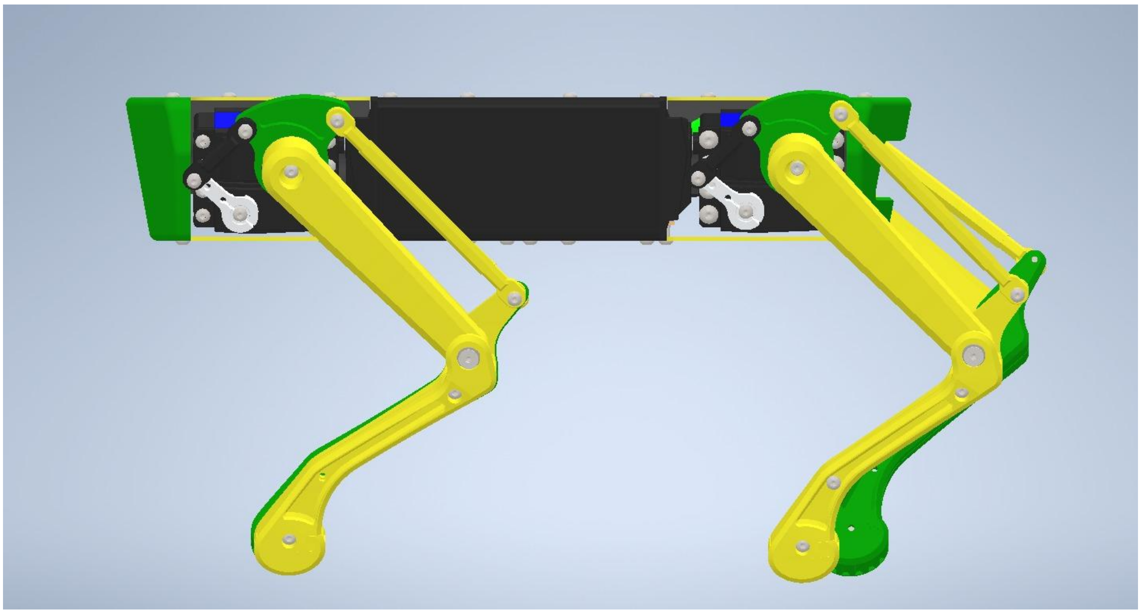


Figure 3.2. Side view of the robot showing overall height and leg geometry

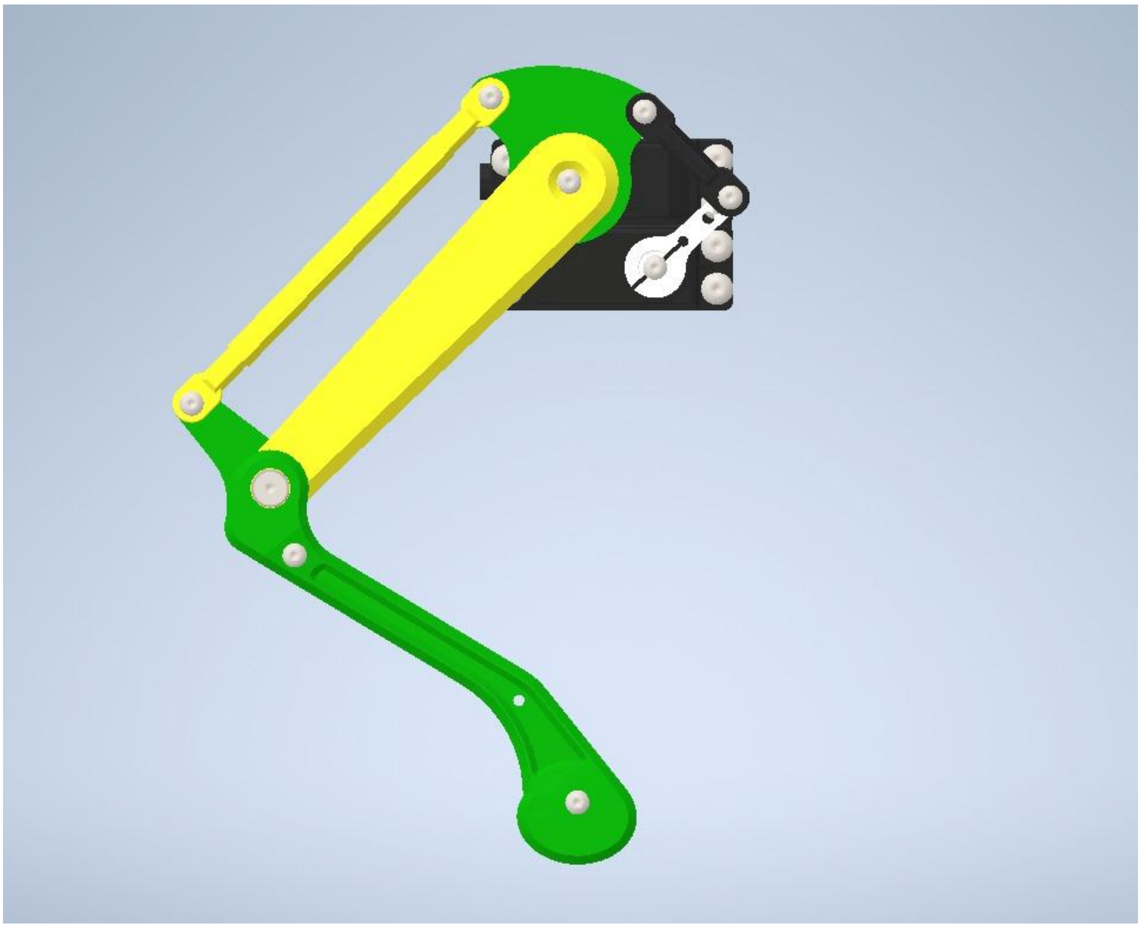


Figure 3.3. Leg segment configuration illustrating coxa, femur, and tibia.

3.3. System Components and Their Selections

The selection of system components was based on the requirements for computations associated with autonomous navigation and artificial intelligence, in addition to the mechanical and electrical requirements of a quadruped robotic platform in a disaster environment. The hardware architecture of this robotic platform has been implemented as a distributed system that separates high-level perception and decision making from low-level motion control, which enhances reliability and permits real-time operation.

The entire system consists of an artificial intelligence processing unit, a real-time motion control unit, sensing modules, actuation components, and a power distribution unit. All components were selected based on their available technical characteristics (e.g., power consumption and compatibility with the overall architecture).

3.3.1. Jetson Nano Orin

NVIDIA Jetson Orin Nano was evaluated as one of the advanced control and perception units of the quadruped robot system due to its high computational performance and optimized architecture for embedded artificial intelligence applications. Its multi-core ARM processor and GPU-accelerated parallel computing capabilities provide significant advantages in real-time vision processing, deep learning–based perception, and state estimation tasks that require intensive computation.

Maintaining stable locomotion in quadruped robots requires sensor feedback data to be processed with minimal latency. Jetson Orin Nano meets this requirement through its high-bandwidth memory interfaces and hardware-accelerated compute units, enabling camera-based obstacle detection, terrain classification, SLAM applications, and proprioceptive sensor fusion to be executed on a single embedded platform.

The platform’s low power consumption provides an advantage for battery-operated mobile robotic systems. The Jetson board will be supplied through a dedicated regulated DC power line to prevent voltage fluctuations and electrical noise from affecting the computing unit. The majority of raw data processing will be executed on the Jetson, enabling Raspberry Pi 5 to focus specifically on real-time servo actuation and balance control tasks.

In conclusion, NVIDIA Jetson Orin Nano is selected as a suitable computing platform for perception, environment awareness, and advanced control functionalities required for quadruped locomotion, owing to its embedded AI computing capability, high-speed parallel processing performance, and suitability for real-time robotic applications.

Figure 3.4. NVIDIA Jetson Orin Nano

3.3.2. Raspberry Pi 5

Raspberry Pi 5 was selected as the central processing unit of the quadruped robot system. Owing to its compact structure, low power consumption, and high processing capability, Raspberry Pi 5 is suitable for real-time control applications. Its advanced multi-core ARM processor allows the execution of complex computation tasks such as gait algorithms, balance control, sensor fusion, and image processing.

Through operating system–level multitasking, coordinated control of robot legs, generation of motor drive signals, and processing of sensor data can be managed efficiently. The processing capacity provided by Raspberry Pi 5 enables stable operation even under demanding computational loads required for quadruped locomotion.

The power requirement of 5 V – 5 A is supplied through an independent regulated power line separated from the main battery system in order to prevent voltage fluctuations from affecting the control electronics. In addition, active cooling modules are integrated to keep processor temperatures within safe operating limits, preventing thermal instability during intensive workloads.

In conclusion, Raspberry Pi 5 is considered an appropriate central controller for the quadruped robot system due to its computing performance, expandability, and suitability for real-time robotic control tasks.

Figure 3.5. Rasperry Pi 5

3.3.3. PCA9685 16-Channel PWM Servo Driver

The PCA9685 16-channel PWM driver module is selected as the servo control interface for the quadruped robot system. Quadruped locomotion requires simultaneous and synchronized actuation of multiple joints; therefore, using a dedicated PWM generator reduces timing uncertainty and prevents excessive workload on the main processor. The PCA9685 includes a 12-bit resolution hardware PWM controller which allows precise control of angular position and motion trajectory for each servo motor.

The module communicates with the main processing unit over the I^2C interface, enabling multiple PCA9685 boards to be addressed and expanded on the same communication line if additional servos are required. This modular architecture supports scalable actuation, especially in biped/quadruped systems where multiple degrees of freedom must be coordinated in real time. The hardware PWM generation of the PCA9685 eliminates jitter associated with software-based PWM signaling and improves stability in closed-loop control applications.

To supply the instantaneous current demanded by multiple servos during dynamic motion phases, an external power line is used for servo power input rather than drawing current from the microcontroller supply. This prevents voltage drops and electrical noise from propagating into the control electronics. The PWM frequency can be configured to match servo specifications and torque response characteristics required for quadruped gait execution.

Based on its multi-channel capability, stable hardware timing, low-latency control characteristics, and compatibility with Raspberry Pi 5 and NVIDIA Jetson platforms, the PCA9685 module is determined to be an appropriate servo driver solution for coordinated leg actuation in the quadruped robotic system.

Figure 3.6. PCA9685 16-Channel PWM Servo Driver

3.3.4. The Arducam 2.3 MP AR0234 Global Shutter Camera

The Arducam 2.3 MP AR0234 global shutter camera module is selected as one of the primary vision sensors for the quadruped robot system. The AR0234 sensor utilizes a global shutter pixel architecture, which enables all pixels in the imaging array to capture light simultaneously. This feature eliminates the rolling-shutter distortions commonly encountered when imaging fast-moving objects or when the camera undergoes rapid motion, both of which are typical conditions for mobile quadruped robots.

With a 2.3-megapixel resolution and a 90° field of view, the camera provides sufficient pixel density and coverage for real-time obstacle detection, terrain classification, and visual odometry applications. The ability to capture low-latency, distortion-free frames is essential for maintaining locomotion stability on uneven ground and for enabling perception-guided motion planning.

The compact structure and flexible ribbon cable interface allow seamless integration with embedded compute platforms such as Raspberry Pi 5 or NVIDIA Jetson Orin Nano. The sensor outputs digital video streams compatible with common high-speed interfaces used in embedded computer vision applications, ensuring deterministic data transfer and reducing synchronization uncertainty in time-critical feedback loops.

By supplying the compute unit with reliable visual information, the AR0234 module plays a critical role in perception, state estimation, and environment awareness. In combination with inertial and proprioceptive sensors, this camera enables robust sensor fusion for autonomous navigation and stable quadruped locomotion.

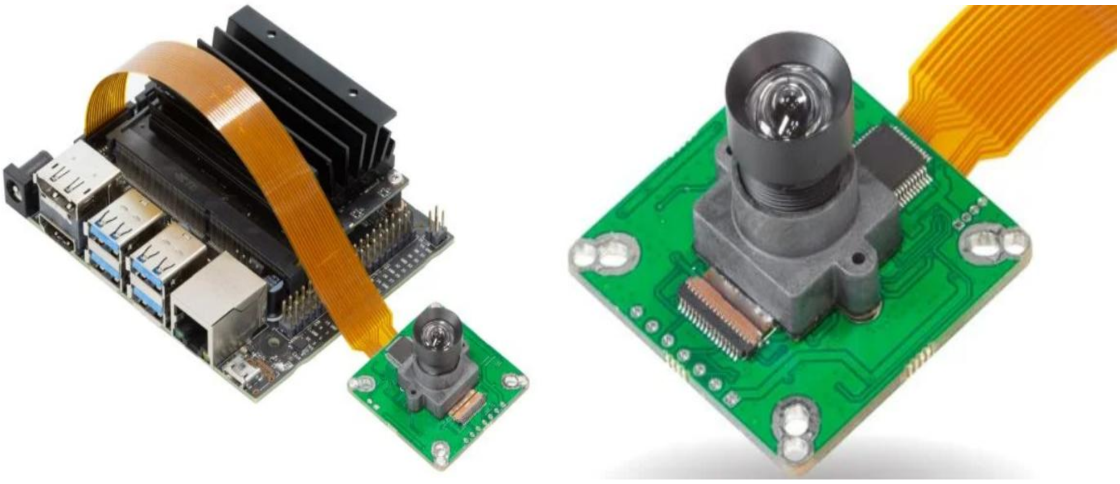


Figure 3.7. The Arducam 2.3 MP AR0234 Global Shutter Camera

3.3.5. RPLIDAR A2M8 360° Lidar Sensor

The RPLIDAR A2M8 lidar unit is selected as the primary distance-sensing subsystem of the quadruped robot platform. The sensor provides a full 360-degree scan range with a maximum detection distance of 12 meters and a sampling frequency of up to 8000 samples per second. These specifications satisfy the real-time perception requirements of a dynamically moving quadruped robot operating in both indoor and semi-structured outdoor environments.

The ability to acquire dense range data enables accurate mapping of the surrounding terrain geometry and reliable obstacle detection during locomotion. The lightweight mechanical structure of the unit minimizes additional mass on the robot body, contributing to lower inertial load during rapid maneuvers and terrain transitions. Its low power consumption makes the device suitable for battery-powered mobile robotic platforms.

The data provided by the lidar are processed within the perception pipeline to generate local occupancy maps and terrain height estimations. These representations are essential inputs for gait stability algorithms and motion planning routines. The angular resolution and rotational scanning mechanism allow the robot to detect discontinuities in the walking surface and adapt its step trajectory to maintain balance in real time.

The sensor communicates with the main processor through a UART or USB interface and provides time-stamped distance measurements suitable for synchronized sensor fusion with IMU and camera outputs. The RPLIDAR A2M8 has full ROS support, enabling rapid integration into higher-level control architectures.

Based on its mechanical robustness, cost efficiency, adequate scanning range, and compatibility with embedded robotic control frameworks, the RPLIDAR A2M8 is considered an appropriate and effective lidar sensor solution for the quadruped robotic platform.

Figure 3.8. RPLIDAR A2M8 360° Lidar Sensor

3.3.6. Positioning and Navigation Unit – u-blox NEO-M8N GNSS Module

The u-blox NEO-M8N GNSS receiver module is selected to support global positioning and navigation tasks within the quadruped robot system. GNSS-based localization enables the robot to estimate its absolute position in outdoor environments, facilitating high-level navigation, waypoint tracking, and environment mapping. The module supports GPS, GLONASS, and QZSS satellite constellations and delivers a typical horizontal positioning accuracy of approximately 2.5 meters under nominal signal conditions.

The NEO-M8N provides a position update rate of up to 10 Hz, which is sufficient to support the locomotion and navigation speeds expected in quadruped robotic platforms. Communication with the onboard processing unit is achieved via a UART interface, ensuring deterministic and low-latency transmission of navigation messages. Support for standard NMEA message formats simplifies data parsing and integration into ROS-based robotics frameworks.

To improve satellite acquisition time and signal robustness, an external active antenna will be employed, and optional SBAS correction capabilities may be enabled depending on deployment conditions. The module’s compact dimensions and low power consumption provide advantages for battery-powered mobile robotics applications where energy efficiency and mechanical integration constraints are critical.

When combined with proprioceptive and exteroceptive sensors such as IMUs and lidar, GNSS data contributes to multi-sensor fusion algorithms that enhance reliability in outdoor navigation. In this context, the NEO-M8N module is considered suitable as the primary GNSS receiver for the quadruped robot platform due to its cost efficiency, adequate positioning performance, and compatibility with embedded control architectures.

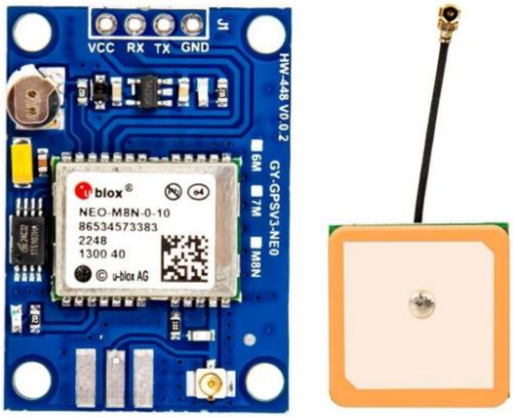


Figure 3.9. Positioning and Navigation Unit – u-blox NEO-M8N GNSS Module

3.3.7. MG996R High-Torque Servo Motor

The MG996R metal-gear servo motor is selected as one of the primary joint actuators in the quadruped robot system. Quadruped locomotion requires high torque levels for joint articulation, particularly during stance phases in which the load of the robot body must be supported by individual legs. The MG996R provides a stall torque capacity of approximately 9–11 kg·cm at 6 V supply voltage and delivers sufficient mechanical power for hip and knee joint actuation within lightweight mobile platforms.

Internal metal gear construction increases mechanical durability and improves resistance to repetitive impact stresses that occur during walking and trotting gait cycles. The servo incorporates an onboard control circuit and potentiometric feedback, enabling closed-loop angular position control through standard PWM modulation. This property facilitates integration with the PCA9685 servo driver, allowing synchronized multi-joint control and stable trajectory execution.

The actuator operates at nominal voltages of 4.8–7.2 V and exhibits current draw levels that increase significantly under load conditions. For this reason, separate regulated power delivery and current limiting protections are included in the power subsystem to maintain system stability and prevent voltage sag during dynamic movements. Shaft angle resolution and torque response characteristics are considered suitable for low-to-medium precision legged locomotion tasks.

In conclusion, the MG996R servo motor is selected based on its high torque output, mechanical robustness, integrated position feedback, and compatibility with the selected servo drive architecture, making it appropriate for joint actuation requirements in the quadruped robotic platform.

Figure 3.10. MG996R High-Torque Servo Motor

3.3.8. 4S 14.8 V LiPo Battery Pack (8000 mAh class)

A 4-cell (4S) 14.8-V Lithium-Polymer (LiPo) battery pack is selected as the primary onboard energy storage unit for the quadruped robotic platform. Quadruped locomotion imposes highly dynamic mechanical loading on leg actuators, resulting in instantaneous current spikes during stance and lift-off phases. LiPo cell chemistry supports high discharge rates and provides the transient current supply necessary for stable servo torque output without excessive voltage droop.

The selected capacity class (approximately 8000 mAh) enables sufficient operational duration for typical experimental locomotion trials, while maintaining a reasonable mass-to-energy ratio suitable for mobile platforms in which battery weight directly influences gait stability and actuator load. The nominal pack voltage of 14.8 V is regulated through dedicated buck converters to supply appropriate voltage rails for the computing units (e.g., Raspberry Pi 5 and NVIDIA Jetson Orin Nano), servo driver circuits, and auxiliary sensors.

LiPo batteries provide high energy density but require careful management of charge-discharge cycles to prevent thermal instability and long-term degradation. For this purpose, over-current, under-voltage, and thermal protection mechanisms are integrated into the battery management subsystem. Cell balancing mechanisms will be applied during charging to prolong usable service life and ensure uniform voltage distribution across series-connected cells.

Based on its high discharge capability, favorable mass-to-capacity characteristics, and compatibility with the dynamic current requirements of multi-servo quadruped actuation, the 4S 14.8-V LiPo battery pack is determined to be an appropriate energy storage component for the proposed robotic system.

Figure 3.11. 4S 14.8 V LiPo Battery Pack (8000 mAh class)

3.3.9.The TP-Link Archer T4U dual-band Wi-Fi adapter

The TP-Link Archer T4U dual-band Wi-Fi adapter is selected to provide wireless communication between the quadruped robot platform and an external control or monitoring workstation. Because the communication architecture does not require cellular connectivity, the Wi-Fi interface enables a cost-efficient and infrastructure-compatible method for transmitting telemetry, sensor status, and control commands through a local wireless network.

The Archer T4U supports IEEE 802.11ac dual-band operation (2.4 GHz and 5 GHz) and connects to the onboard computing system through a USB 3.0 interface. The 5 GHz band enables higher throughput and reduced latency, which are critical for real-time communication tasks such as state updates, operator command input, and low-bandwidth video streaming. Linux driver compatibility facilitates seamless integration with the Jetson platform, allowing the network interface to be recognized as a standard wireless adapter by the operating system.

Figure 3.12. The TP-Link Archer T4U dual-band Wi-Fi adapter

3.4. Applied Methods

This section describes the engineering methods applied during the design of the quadruped robot, including mechanical load analysis, actuator torque estimation, kinematic modeling, power analysis, and control strategy selection. These methods are used to guide component selection and system architecture design.

3.4.1. Load Distribution and Static Analysis

To estimate the mechanical load acting on each leg, the total robot weight is assumed to be evenly distributed among four legs during static standing conditions. The load acting on a single leg is calculated using **Equation (1)**.

$$\boldsymbol{F_{leg} = \frac{W}{4}} \qquad (1)$$

W is the total robot weight and F_{leg} is the load acting on a single leg.

For a robot mass of 3.5 kg, the force acting on each leg is calculated as given in **Equation (2)**.

$$\boldsymbol{F_{leg} = \frac{3.5 \times 9.81}{4} \approx 8.6\ N} \qquad (2)$$

3.4.2. Servo Motor Torque Calculation

Servo motors are selected based on the torque required to support the robot body and generate leg motion. The required torque for a joint is estimated using **Equation (3)**.

$$\boldsymbol{\tau = F \cdot r} \tag{3}$$

Using an effective lever arm length of approximately 0.12 m, the required torque per joint is calculated as shown in **Equation (4)**.

$$\boldsymbol{\tau \approx 8.6 \times 0.12 = 1.03\ \mathrm{Nm}} \tag{4}$$

A safety factor greater than 1.5 is considered to account for dynamic loads during locomotion, leading to the selection of 20 kg·cm torque servo motors.

3.4.3. Structural Strength of 3D Printed Parts

The mechanical strength of the 3D-printed leg and body components is evaluated using bending stress analysis. The bending stress is calculated according to **Equation (5)**.

$$\boldsymbol{\sigma = \frac{M \times c}{I}} \tag{5}$$

σ is bending stress,

M is bending moment,

C is the distance from the neutral axis, and

I is the area moment of inertia.

This analysis guides the selection of wall thickness, infill ratio, and filament material to ensure sufficient strength while minimizing weight.

3.4.4. Kinematic Modeling and Inverse Kinematics

Leg motion is controlled using inverse kinematics to determine the joint angles required to achieve a desired foot position. For a planar two-link leg model, the joint angles are calculated using **Equations (6) and (7)**.

$$\boldsymbol{\theta_2 = \cos^{-1}\left(\frac{x^2 + y^2 - l_1^2 - l_2^2}{2 l_1 l_2}\right)} \tag{6}$$

$$\boldsymbol{\theta_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{l_2 \sin\theta_2}{l_1 + l_2 \cos\theta_2}} \tag{7}$$

where $l_1$and $l_2$represent the femur and tibia lengths.

3.4.5. Power Consumption and Battery Capacity Estimation

The total current consumption of the robot is estimated by summing the current demands of major components as expressed in **Equation (8)**.

$$I_{total} = \sum I_{servos} + I_{processors} + I_{sensors} \qquad (8)$$

The approximate operating time of the battery is calculated using **Equation (9)**.

$$T = \frac{C}{I_{total}} \qquad (9)$$

C is the battery capacity in ampere-hours.

This estimation supports the selection of a 3S Li-Po battery to meet operational requirements.

3.5. Software Used

The software architecture of the quadruped robot is developed using a layered approach separating high-level perception from low-level motion control.

The main software tools and platforms used include:

- **Ubuntu Linux** running in **VirtualBox** for development
- **ROS2** for inter-process communication and modular software design
- **Gazebo** for simulation and visualization
- **Raspberry Pi Imager** for operating system deployment

High-level tasks such as image processing and object detection are executed on the NVIDIA Jetson Nano, while real-time control tasks including inverse kinematics and servo control are handled by the Raspberry Pi.

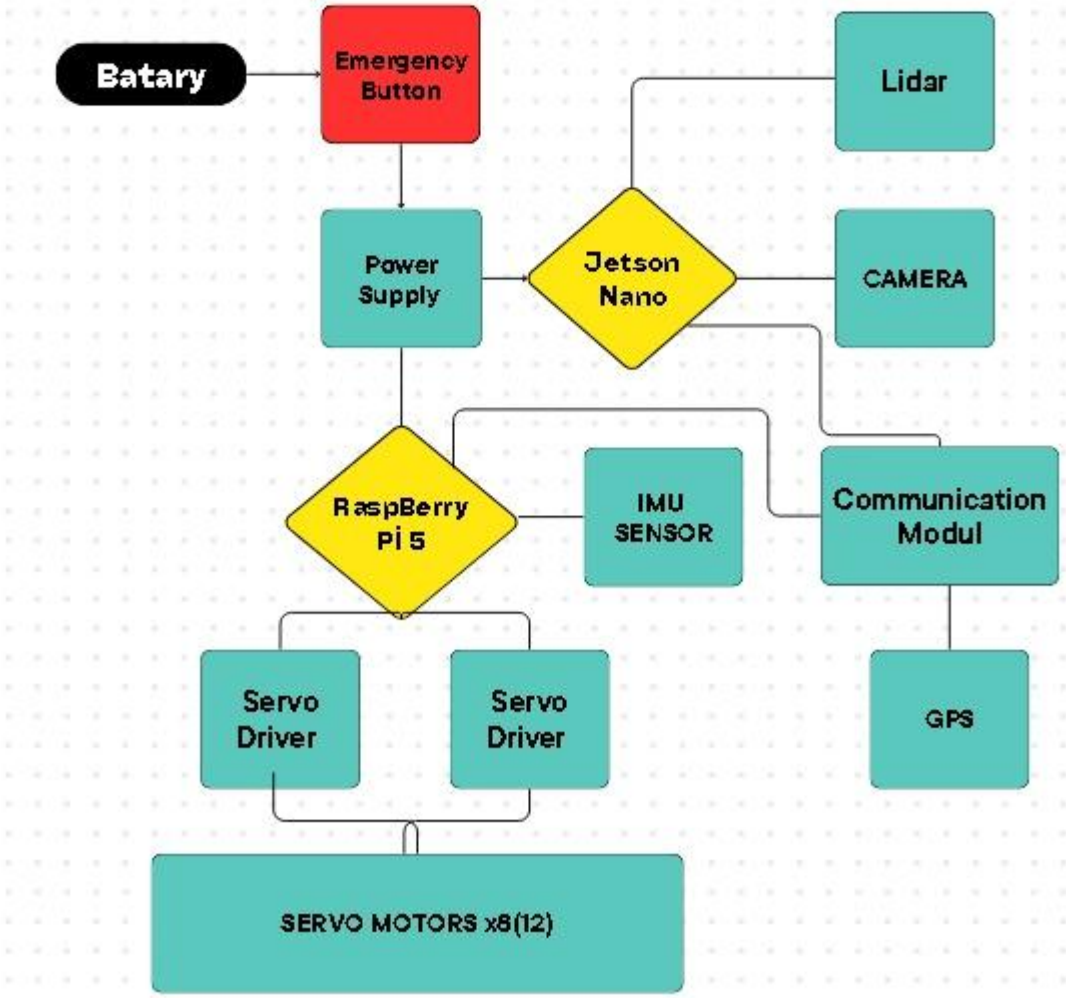


Figure 3.13. Software architecture and data flow diagram.

3.6. Component List and Economic Analysis

After the completion of the project design phase, the proposed system is planned to be realized through the procurement and integration of the required hardware components. Market research was conducted for all materials specified in the design stage of the quadruped robot, and a preliminary cost calculation was performed based on current market prices. The component list and the estimated cost analysis resulting from this planning are presented in Table 3.1.

Table 3.1. Component List

Component Name	Purpose of use	Unit Price (TL)	Quantity	Price (TL)
Waveshare Jetson Orin NX 16GB AI Kit (WiFi)	High-level AI processing, image processing, object detection	52,489.32 TL	1	52,489.32 TL
Raspberry Pi 5	Low-level motion control and inverse kinematics calculations	3,015.35 TL	1	3,015.35 TL
PCA9685 16-Channel I2C PWM Servo Driver	PWM signal generation for servo motors	195.71 TL	1	195.71 TL
MG996R Metal Gear Digital Servo Motor	Leg joint actuation	172.46 TL	8	1,379.68 TL
TP-Link Archer TX20U Plus	Wireless communication (WiFi)	1,501.48 TL	1	1,501.48 TL
Arducam 2.3MP AR0234	Visual perception and camera input	4,686 TL	1	4,686 TL
A2M8 RPLIDAR 360°	Distance measurement & obstacle detection	11,842.62 TL	1	11,842.62 TL
Tinylab 3D 1.75 mm ABS Filament	Mechanical part manufacturing	3,040.00 TL	1	3,040.00 TL
			TOTAL	78,150.16 TL

3.7. Legal Aspects

This section evaluates the potential legal and regulatory considerations related to the development and use of the proposed quadruped search and rescue robot. Since the robot is designed for operation in disaster and emergency environments, compliance with relevant safety, electrical, and operational regulations is taken into account during the design phase.

- **Safety for Batteries and Electricity**

The robot operates with LiPo batteries as the main power source. Because LiPo batteries can present dangers related to overheating, short-circuiting, and damage from impact, the design implements a means of providing voltage regulation and current protections. The entire electrical system meets the general principles of safety for low-voltage equipment to limit the level of danger throughout the usage of the robot.

- **Mechanical Safety**

Movement via removable parts, such as joints and mechanism, is part of the structure of the machine. When in action, these mechanics can create a hazard for users through possible injury from moving parts. To reduce this risk, cutting areas were not included in the design of the mechanism and were positioned away from any areas where direct access could occur. The intended environment for using this particular piece of machinery is in emergency response situations, where trained personnel will operate it within a controlled setting.

- **Operational Use in Disaster Areas**

Structurally, the system has been created to serve primarily as an aid to humanitarian agencies conducting search and rescue missions rather than as a substitute for human input. Furthermore, its deployment in areas affected by disasters will require the appropriate approvals by local public administrations/authorities, as well as governmental/emergency response organizations. The software features and attributes of this robotic system are consistent with those used in traditional operational guidelines established for emergency response operations.

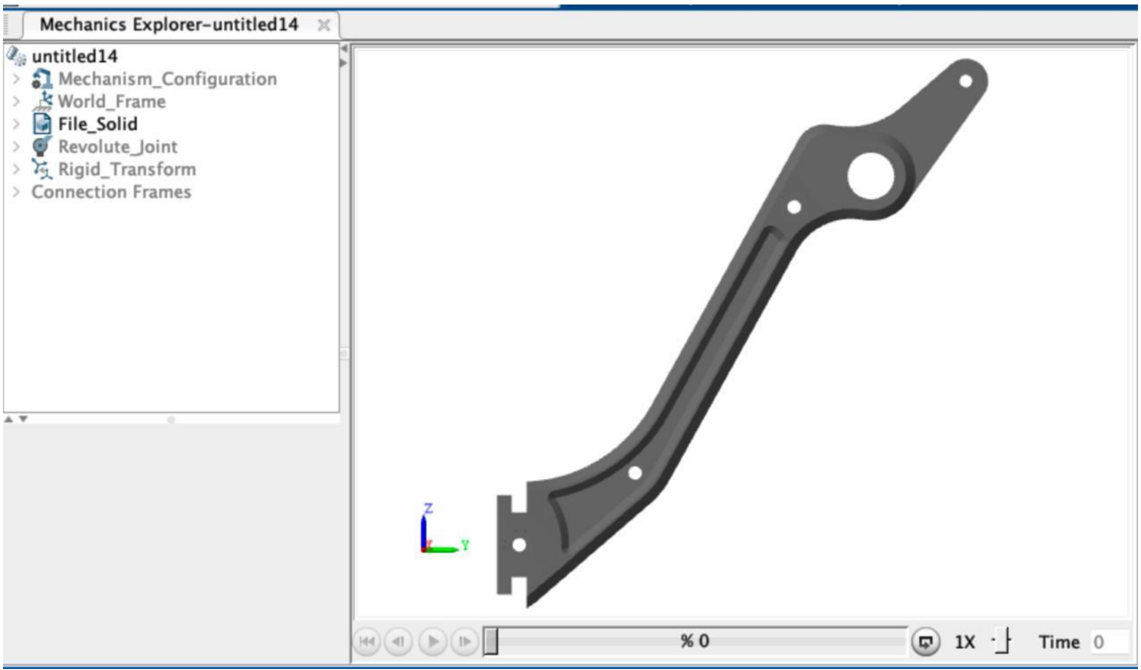
- **Data Privacy and Ethical Considerations**

The robot uses camera and sensor systems to perceive its environment. Image and sensor data collected during operation are intended solely for search and rescue purposes. Personal data protection principles are considered, and data usage is limited to operational requirements. Any future use of the system involving data recording or transmission would need to comply with applicable data protection regulations.

4. SIMULATION STUDIES

4.1. General Information

The design of the robotic leg mechanism is bio-inspired and follows a typical quadruped animal model defined by a skeleton. The design is comprised of two main parts: Upper Leg (Femur) and Lower Leg (Tibia).The Upper Leg is a primary support link for the chassis and also serves as a connection to both the chassis and Lower Leg (Tibia). The Tibia describes the bottom section (lower leg) of the robotic leg and provides contact with ground. The ends of these parts are referred to as end-effectors. Lastly, the physical dimensions of the parts were determined based on dimensionally accurate 3D printed replicas of both parts: "femur v2_4pcs.STL" (upper leg) and "tibia v4_R-onepiece.STL" (lower leg). The dimensions of the links (i.e. L_1 and L_2) are set to 100 mm in the simulation to match proportions of the physical prototype.



. **Figure 4.1.** 3D Visualization of the Tibia Component

3D view of the robot’s tibia segment within the Simscape Multibody environment. The part's center of inertia and joint orientation have been computationally verified in the digital domain to ensure physical accuracy during simulation.

4.2. Simulation Software

The dynamic performance and kinematic precision of the design were evaluated using the Simscape Multibody environment in MATLAB/Simulink. In this simulation setup, there were three primary blocks: The World Frame and Solver Configuration define the global coordinate frame and the parameters for numerical integration; The Mechanism Configuration specifies the acceleration due to gravity of -9.81m/s^2, which simulates a physical environment with this force; and The Revolute Joint represents the hip and knee joints, which provide these joints with the ability to rotate about an axis of rotation. The File Solid block imports the STL geometries, with all associated mass and inertia values specified in the physics engine.

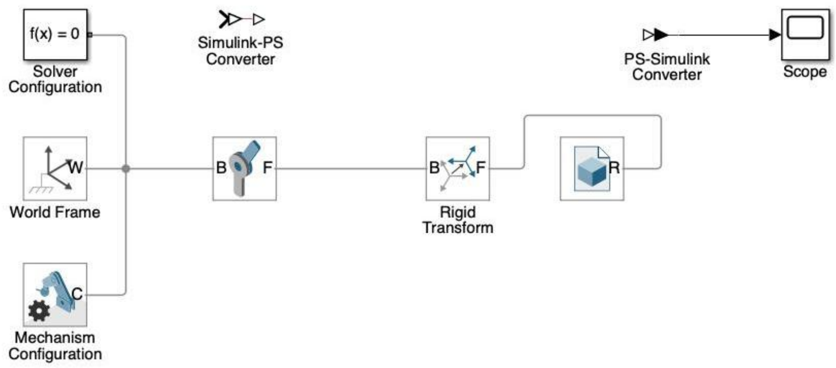


Figure 4.2. Simulink Block Diagram of the Robotic Leg Mechanism.

Simulink Block Diagram of the Robotic Leg Mechanism:

This diagram illustrates the integrated simulation architecture where mathematical control algorithms and physical multi-body dynamics are coupled to analyze the leg's kinematic behavior.

4.2. System Modeling

Inverse Kinematics (IK) is the process of calculating the required joint angles to move the foot-end to a specific target (x, y) coordinate. The model relies on the Law of Cosines to solve the geometric relationship between the links.

The the joint angle (q) is calculated by using **Equation (10).**

$$\boldsymbol{D} = (\boldsymbol{x}^2 + \boldsymbol{y}^2 - \boldsymbol{L1}^2 - \boldsymbol{L2}^2)/(\mathbf{2} * \boldsymbol{L1} * \boldsymbol{L2}) \qquad (\mathbf{10})$$

q = arccos(D)

This ensures that for any reachable target point, the system generates a stable angular command for the actuator.

4.3. Simulation

Algorithm Implementation

In the preceding section, we implemented the mathematical model via a MATLAB function block within a Simulink environment. Using this implementation, the target Cartesian coordinates (x, y) can be converted into joint angles (q). By doing this, we can ensure that the robot properly reaches the desired touchdown points within its gait cycle.

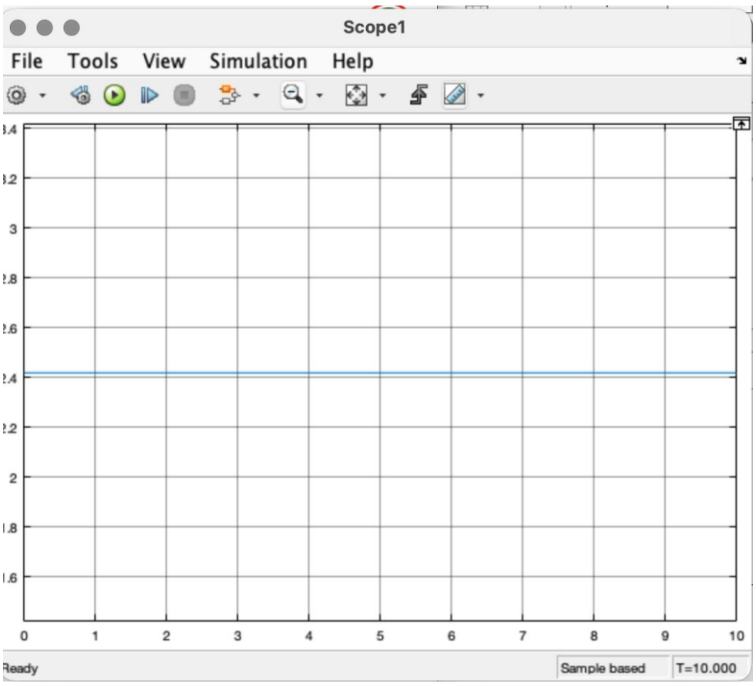


Figure 4.3. The MATLAB Function block that implemented the inverse kinematics algorithm can be seen in conjunction with the Scope output from constant (x,y) inputs. It was able to compute joint angles of 2.4 radians corresponding to the target point and achieve a consistent mechanical position.

4.4. Dynamic Testing and Trajectory Tracking

The inverse kinematics (IK) algorithm controlled the leg mechanism in response to dynamic input signals. In order to maintain the numerical stability of these signal inputs during testing, all inputs had second-order filtering applied to them prior to being input into the IK algorithm. Likewise, the Simscape engine performed automatic computation for joint torque values throughout the entire testing process resulting in successfully tracking of the targeted trajectories by the foot-end.

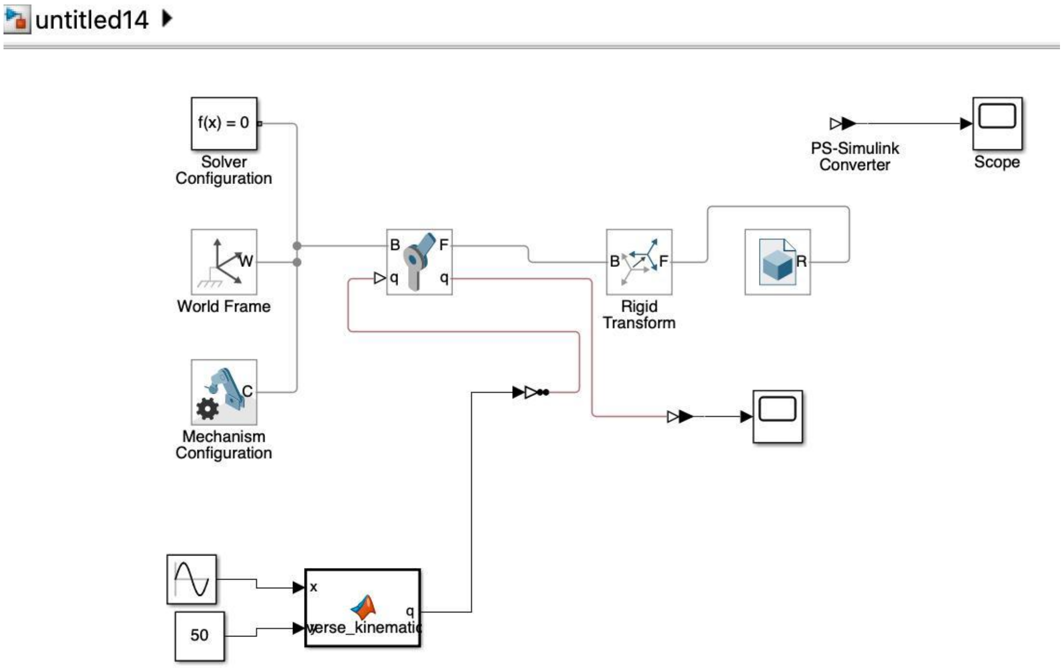


Figure 4.4. Dynamic Control Block Diagram shows the conceptual overview of the Dynamic Control System Architecture. The Figure depicts the process of sending the Reference Sine Wave from MATLAB Function Block (MFB) to Joint Actuator.

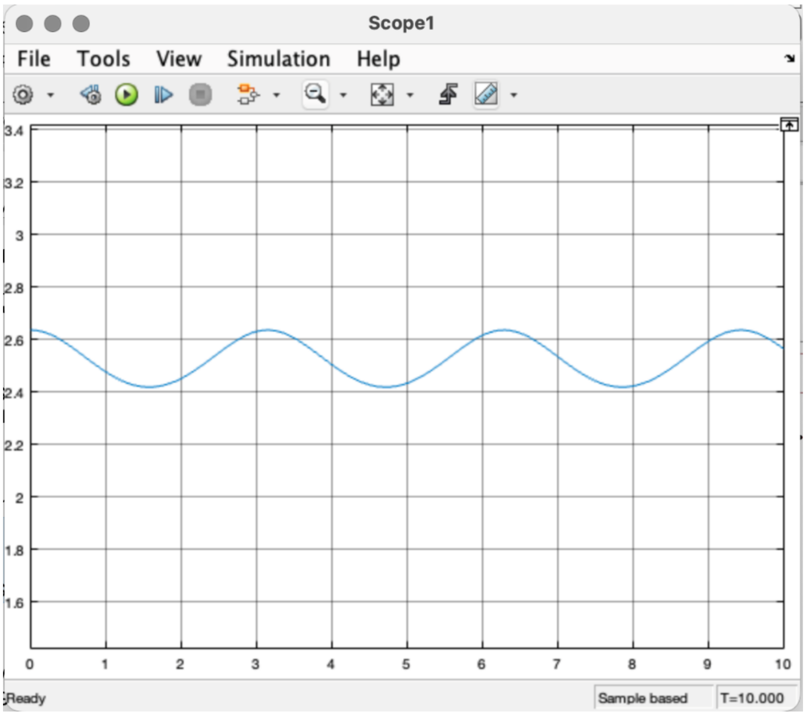


Figure 4.5. Dynamic Trajectory Tracking Graph shows the Trajectory Tracking and Periodic Output graph, which records a periodic (sinusoidal) output as measured with the Scope, and illustrates how well the Trajectory Tracking will perform and the periodic motion of the foot during the Gait Cycle.

5. RESULTS

5.1. General Information

The technical products, numerical information, and visual representation of results contained in this section show the success (or failure) of meeting the stated goals of the search and rescue robot project. The academic community does not accept photographs of an experiment as a valid result; rather, graphs, figures, and tables provide tangible evidence of the system achieving its goal.

The results of this study are separated main groups: simulation results provided by computer -generated models; and hardware experimental results based upon tests conducted on a prototype constructed from physical materia

5.2. Simulation Results

Using the Simscape Multibody environment in MATLAB/Simulink, a simulation model of a quadruped robot was created to analyze kinematic accuracy and trajectory tracking.

Inverse Kinematics (IK) Accuracy: A mathematical model based on the Law of Cosines was implemented as MATLAB Function Block to convert Cartesian coordinates into joint angles.

Target Achievement: When the algorithms were tested with constant (x,y) inputs, the algorithm calculated joint angles of 2.4 radians, allowing the robotic leg to achieve one consistent mechanical position. Trajectory Tracking: Through dynamic control tests, using a sine wave reference command to the joint actuators allowed simulating the gait cycle.

Gait Cycle Stability: The graphs derived from trajectory tracking tests illustrated the ability of the foot-end to follow periodic motion throughout the locomotion phases of stability.

5.3. Experimental Results

The project was physically built using a combination of high-level perception and low-level control, which was accomplished by using a hybrid processor architecture.

Actuation and Control: The Raspberry Pi 5 received PWM signals with 12-bit resolution from the PCA9685 driver, allowing the application of 12-bit PWM signals to provide 4096 steps of precise positioning of the metal gear servo motors.

Edge AI Performance: The Jetson Nano allowed for an edge-based deep learning model (YOLO) to locally execute the detection of human body shapes and "thermal signals of life" from camera feeds.

Mapping and SLAM: By combining 360-degree scanning information (from an RPLIDAR A2M8) with orientation data from the Inertial Measurement Unit (IMU), the robot built an Occupancy Grid (OG) map with accurate positional coordinates.

Environmental Perception: The use of a global shutter camera (Arducam) removed rolling shutter effects that occur when moving rapidly so that accurate visual information could be gathered for avoidance of obstacles.

Power Management: The 14.8V 8000mAh lithium polymer (LiPo) battery had enough available amps to supply power to both processor and actuation layers, thus permitting maximum search and rescue mission duration.

5.4. Comparative and Economic Analysis

The findings show that this venture represents a notable departure from previously established solutions within functioning businesses. Cost Effectiveness – The price of commercial-based platforms such as Boston Dynamics' Spot robot can be anywhere between $70k to $150k (USD). Our prototype replicates the essential capabilities of these systems (including autonomy and AI) at roughly 5% of their total costs.

Mechanical Robustness – Our system utilized standard servo motors rather than professional-grade actuators, given that we calculated a safety factor >1.5(1) for (A) each joint with respect to (B) the torque rating/system capacity (i.e., 1.03Nm) when applied along the axis of rotation during dynamic load conditions inherent to gait or locomotion.

Operational Sustainment – Due to the use of 3D printing technology (also known as Additive Manufacturing), broken/exhausted limb components can now be changed or replaced anywhere in the world with minimal time and effort as they were designed to allow for “Field Repair” operation; consequently, components could be provided to users without needing to ship them using conventional distribution infrastructures/logistics systems.

6. CONCLUSİONS

6.1. Evaluation of the Study

The main aim of this study was to create a Quadruple Robot Dog for Search and Rescue Missions that can operate effectively in disaster zones where standard wheeled or tracked machines typically do not perform well. This Project integrated mechanical design, electronics hardware and embedded artificial intelligence systems into a working prototype that can move and hold its balance on unstable surfaces. Furthermore, by concentrating on the "Golden Hours" aspect of this venture, this Research provides an answer to how to rapidly mitigate dangerous conditions for those trapped underneath debris, thus allowing for faster and more accurate assessments of sites.

6.2. Future Work

Although the present prototype currently meets the basic operational requirements for autonomous motion and life detection, there are many opportunities for improvements in the following areas:

Control Algorithms: The use of PID based balanced control systems can be advanced to Model Predictive Control (MPC) for improved performance in handling more complex dynamic environments, as well as not overloading the available processing units.

Navigation and Communication: Further improvements to 4G modem interfacing and GPS waypoint tracking will provide a larger operating range of decentralized disaster zones.

Hardware Upgrade: Upgrading from typical hobby servos to a quasi-direct drive actuator system may provide a greater torque density and mechanical compliance as do many higher-end research and academic platforms such as the Mini Cheetah.

6.3. Problem Resolution and Process Facilitation

This research also fills a gap between very expensive industrial robots and low autonomy hobbyist systems in terms of functionality/style/cost.

Cost Savings - The prototype shows an example of an efficient and effective search and rescue device that can be manufactured 95% less than industrial devices such as Boston Dynamics' Spot.

Field Sustainability - Using 3D printing technologies to manufacture the device will provide "field repairability" by allowing for immediate replacement of broken limbs without creating a complex logistical supply chain.

Safety - The robot helps with "vanguard reconnaissance" meaning that it can go ahead of humans into dangerous buildings or gas leak tunnels before the humans do, thus greatly increasing worker safety.

6.4. Target Customers and Application Areas

The prototype created in this research can be commercially beneficial in a variety of fields.

Local governments and other public organizations (such as AFAD - Turkish Emergency Services - or State Fire Departments) will be able to lessen their dependence on foreign manufactured goods by using these robots.

The prototype has applicability in the Mining Industry and for Industrial Inspection where there are places that may be considered unsafe or out of the reach of individuals.

The modular design of this prototype is advantageous in that it can be adapted for many different types of Non-Civilian Reconnaissance roles in the defense industry.

6.5. Final Budget Evaluation

The final total budget for the development of the prototype came to 78,150.16 Turkish Lira (TL). While much of the expense was attributed to high-performance components, specifically the NVIDIA Jetson Orin Nano and RPLIDAR A2M8, the full cost of developing this prototype is still considerably less than that of an industrial-grade search and rescue (SAR) platform. As such, it is confirmed that the project met its goal to provide 'Accessible Technology'.

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APPENDICES

ANNEX - 1. IEEE Code of Ethics (English and Turkish)

IEEE Code of Ethics (English)

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. To accept responsibility in making decisions consistent with the safety, health, and welfare of the public, and to disclose promptly factors that might endanger the public or the environment.
2. To avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist.
3. To be honest and realistic in stating claims or estimates based on available data.
4. To reject bribery in all its forms.
5. To improve the understanding of technology, its appropriate application, and potential consequences.
6. To maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience.
7. To seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others.
8. To treat all persons fairly and with respect, without discrimination based on race, religion, gender, disability, age, or national origin.
9. To avoid injuring others, their property, reputation, or employment by false or malicious action.

10. To assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

IEEE Etik Kuralları (Türkçe)

IEEE üyeleri olarak, teknolojilerimizin dünya genelinde yaşam kalitesi üzerindeki etkisinin bilincinde olarak ve mesleğimize, meslektaşlarımıza ve hizmet ettiğimiz topluma karşı kişisel sorumluluğumuzu kabul ederek, en yüksek etik ve profesyonel davranış ilkelerine bağlı kalmayı taahhüt ederiz:

1. Kamu güvenliği, sağlığı ve refahı ile uyumlu kararlar almaktan sorumlu olmak ve kamuya veya çevreye zarar verebilecek unsurları zamanında açıklamak.

2. Gerçek veya algılanan çıkar çatışmalarından mümkün olduğunca kaçınmak ve mevcut olduğunda ilgili tarafları bilgilendirmek.

3. Mevcut verilere dayalı olarak iddia ve tahminlerde dürüst ve gerçekçi olmak.

4. Her türlü rüşveti reddetmek.

5. Teknolojinin anlaşılmasını, uygun kullanımını ve olası sonuçlarını geliştirmek.

6. Teknik yeterliliği sürdürmek ve geliştirmek; yalnızca eğitim ve deneyimle yetkin olunan konularda görev üstlenmek.

7. Teknik çalışmalara yönelik dürüst eleştirilerde bulunmak, hataları kabul edip düzeltmek ve başkalarının katkılarını doğru şekilde belirtmek.

8. Irk, din, cinsiyet, engellilik, yaş veya milliyet ayrımı yapmaksızın tüm bireylere adil ve saygılı davranmak.

9. Yanlış veya kötü niyetli eylemlerle başkalarına, mallarına, itibarlarına veya işlerine zarar vermekten kaçınmak.

10. Meslektaşların profesyonel gelişimine destek olmak ve bu etik kurallara uymalarını teşvik etmek.

ANNEX - 2. Student Ethic Form

OSTIM TECHNICAL UNIVERSITY ANKARA

FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION PROJECT

DECLERATION

Ethical Behavior and Professional Honesty

Fair and accurate assessment are key factors that must be at the core of education. Any act of a student that would cause unfair advantage to himself/herself or another student is considered an academic dishonesty. It is also academic dishonesty to copy someone else's work and make it look like his/her own. This is called theft.

Allowing someone else to copy your work and make it look like his/her own is also fraud. It is both wrong and unethical to provide all kinds of unfair profits and advantages. It is also wrong to prepare the assignments, projects or laboratory reports of the other students. Doing this is not ethical, neither. Likewise, having his/her own homework, project or laboratory report being prepared and written by someone else is also a complete deception and immorality.

Using another student's answers, homework or reports without his/her knowledge is nothing but stealing that student's labor and this is called theft. It is the same thing to pass the course by cheating in exams.

We should not forget that those who cannot learn the truth when they are students continue these mistakes after graduation and become one of the characters we complain about in the society.

We expect our students and alumni to have ethical and humanly behaviors during both their professional business and social lives. For this, we want you to promise us that you will obey the ethical rules by signing this form.

Let us fight with unethical behavior together.

Asst.Prof.Dr. Şenol GÜLGÖNÜL
Supervisor

I believe what is written above is correct within the framework of ethical rules. I promise to behave ethically during and after my education.

Student No	Student Name and Surname	Date	Signature
210202015	Sıla ÖZDEMİR	19/12/2025	
210202009	Berna Meltem YILDIRIM	19/12/2025	
210202003	Ahmet Emirhan GÖKTÜRK	19/12/2025	
210202023	Uygar BAŞ	19/12/2025	

Each student must fill out this form with a blue pen, sign it, and add it to Graduation Project file as an Appendix

ANNEX - 3. Restrictions (Constraints) Form

Constraint Category	**Explanations**
Design Constraints	During the mechanical and software design, dimensional limits of the robot (300×120 mm body limits), structural stability, center of mass distribution, torque limits of MG996R servos, communication protocol limitations (I2C/UART) and ROS integration requirements impose constraints. EMC/electromagnetic safety and isolation issues must be considered because Li-Po batteries, high-current servo drivers and Jetson Nano are used in close proximity.
Prototype Manufacturing Constraints	The prototype is manufactured using 3D printing. Strength limitations of ABS/PLA materials, layer adhesion, maximum part size of the printer, and tolerance limitations affect final part dimensions. Supply of electronic components, heat dissipation of control boards and EMI shielding are considered.
Implementation / Operational Constraints	In field deployment, the constraints include energy limits, maintenance frequency, servo overheating, wireless latency, and signal loss within disaster debris. Sudden power loss and risk of overcurrent requires safe shutdown procedures.
Social Impact Constraints	The robot contributes to search and rescue operations before human entry into hazardous areas; it should not, however, impede field teams, being mindful of the AFAD/municipality emergency workflow, nor create false victim signals that mislead rescue efforts.
Health Constraints	The system design shall not threaten personnel. Rotating joints and servos require mechanical shielding. Li-Po battery requires safe charging/discharging management to prevent explosion risk. Wireless communication shall not exceed the safe RF exposure limits for operators.
Environmental Constraints	The prototype shall prevent material leakages, heat damages, or environmental contaminations. Plastic parts, batteries, lubricants, and electronic wastes shall be disposed of following environmental regulations when the prototype's lifetime is complete.
Safety Constraints	Emergency stopping, short-circuit protection, safe cut-off of the battery, over-temperature protection, and proper grounding are very important. In printed parts, sharp edges must not be used. During operation, adequate safe approach distance must be maintained between the robot and operator.
Financial Constraints	Component selection was mostly constrained by budget. No industrial motors or sensors could be used and low-cost versions had to be picked instead, such as MG996R servos and a PCA9685 PWM module.
Legal Constraints	Use of wireless communication modules must comply with national radio frequency regulations. Sensor processing and images collected

	must comply with KVKK/GDPR personal data privacy laws. Field use requires permissions from emergency authorities.

ANNEX - 4. Interdisciplinary Work

Interdisciplinary Study	Explanations
Workshops or courses organized by the Head of the Department	During this project work, experiences were also gained in interdisciplinary workshops or courses organized by the Head of the Department and compulsory to attend. Throughout the semester, interdisciplinary seminars on engineering ethics, project management, and problem-solving methodologies were attended within the frame of compulsory interdisciplinary studies. These studies contributed to establishing communication across engineering disciplines and improved systematic decision-making and teamwork skills. Further, some studies for the Graduation Project necessitated collaboration outside the department. Within the framework of the needs related to the mechanical design of the quadruped robot, Volkan Yaldız, a Mechanical Engineering graduate from Çankaya University, was reached for support related to Solid modeling and the principles of mechanical design. Around 12 hours of consultancy were devoted to the work on CAD modeling, selection of mechanical reinforcement, and tolerance analysis for 3D-printed pieces. Additionally, the project team members attended online technical seminars, viewed Solid design tutorials, and engaged in video-based training resources to enhance interdisciplinary competencies. The out-of-department interactions greatly contributed to completing the robot chassis and leg structure according to the mechanical design constraints.

ANNEX - 5. Other Appendices

A5.1. Inverse Kinematics Software Code

This appendix presents the inverse kinematics function used for calculating the joint angle of a quadruped robot leg. The algorithm is based on a two-link leg model consisting of femur and tibia segments. The law of cosines is used to determine the joint angle required for a given target position in Cartesian coordinates.

Code Snippet 4.1. Inverse Kinematics Function

Function q = inverse_kinematics (x, y)

% L1 and L2 represent the leg segment lengths (in mm)

% L1: Femur length

% L2: Tibia length

L1 = 100;

L2 = 100;

% Inverse Kinematics Calculation (Result in Radians)

% This formula determines the required joint angle

% for a target (x, y) position

$$\boldsymbol{D = (x^2 + y^2 - L1^2 - L2^2)/(2 * L1 * L2)} \qquad (\mathbf{10})$$

% Constraint check to prevent complex values

q = acos(max(min(D, 1), -1));

end

ANNEX - 6. Short Biographies

ANNEX 6.1. Berna Meltem YILDIRIM

Berna Meltem Yıldırım was born on 30 July 2002 in Ankara, Türkiye. She completed her primary, secondary, and high school education in Ankara. She is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.2. Sıla ÖZDEMİR

Sıla Özdemir was born on 26 July 2002 in Ankara, Türkiye. She completed her primary, secondary, and high school education in Ankara. She is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.3. Uygar BAŞ

Uygar Baş was born on 16 March 2002 in Ankara, Türkiye. He completed his primary, secondary, and high school education in Ankara. He is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX 6.4. Ahmet Emirhan GÖKTÜRK

Ahmet Emirhan Göktürk was born on 01 March 2002 in Ankara, Türkiye. He completed his primary, secondary, and high school education in Ankara. He is currently a fourth-year student in the Department of Electrical and Electronics Engineering at Ostim Technical University.

ANNEX - 7. Check list of the submission requirements for Graduation Project I

OSTIM TECHNICAL UNIVERSITY ANKARA

FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION PROJECT – I EVALUATION

GRADUATION PROJECT I - FILE SUBMISSION CHECKLIST

STUDENT INFORMATION DATE: 19 / 12 / 2025

Number	Name and Last name	Signature
210202009	Berna Meltem YIIDIRIM	
210202015	Sıla ÖZDEMİR	Sıla Öz…
210202003	Ahmet Emirhan GÖKTÜRK	
210202023	Uygar BAŞ	

This form will be filled out and signed according to the number of students in the project. Students who sign and submit this file accept the accuracy of the information they provide below.

PROJECT INFORMATION

Project Title	:	Design and Implementation of a Search and Rescue Robot Dog for
Project Supervisor	:	Asst. Prof. Dr.Şenol GÜLGÖNÜ Signature

FOR THE PROJECT FILE TO BE SUBMITTED

All of the following answers must be YES.

1. Is the number of verbatim quotations from any source in your dossier LESS than 20% and is the number of paragraphs of verbatim quotations zero? YES ☑ NO ☐
2. Does your dossier include a preface, abstract, main sections (Introduction, Theoretical Background, Design, Simulation, Experimental Study, Results, Evaluations) and a list of references? YES ☑ NO ☐
3. Is there an Introduction, Literature Review, Originality (novelty), Methodology, Impact, List of Standards and Work Schedule in the Introduction section? Is there leadership sharing and B plans in the work packages? YES ☑ NO ☐
4. Has a financial analysis commentary and a legal assessment that may arise from the project been added at the end of the design section? YES ☑ NO ☐
5. Are all equations numbered, all figures captioned and all charts captioned? YES ☑ NO ☐
6. Are all equations, figures, charts and references cited in the text? YES ☑ NO ☐
7. Are all the figures in the Design, Simulation and results section yours? YES ☑ NO ☐
8. Are all references listed in accordance with the Graduation Project Writing Guidelines and at least 5 scientific articles, theses and dissertations are cited? YES ☑ NO ☐
9. Have you included the IEEE Code of Ethics in the Appendix? YES ☑ NO ☐
10. Have you conducted interdisciplinary work and explained this in the Appendix? YES ☑ NO ☐

In order for the project file to be delivered, all of the above answers must be YES.

ANNEX - 8. Graduation Project Student Tracking Form

OSTIM TECHNICAL UNIVERSITY ANKARA

FACULTY OF ENGINEERING
ELECTRICAL & ELECTRONICS ENGINEERING

GRADUATION PROJECT

Student	Number, Name and Surname	:	1. 210202009, Berna Meltem YILDIRIM
		:	2. 210202003, Ahmet Emirhan GÖKTÜRK
		:	3. 210202015, Sıla ÖZDEMİR
		:	4. 210202023, Uygar BAŞ
	Project title	:	Design and Implementation of a Search and Rescue Robot Dog for Disaster Areas
Supervisor	Title, Name and Surname	:	Asst. Prof. Dr. Şenol GÜLGÖNÜL

STUDENT WORK TRACKING SHEET **Education Year** 2025 / 2026

Date visited	Status of the Graduation Project	Advisor's signature
Week 1	Literature review on quadruped robots and search and rescue systems.	
Week 2	Investigation of existing robot dog platform.	
Week 3	System architecture was determined. Main hardware components and sensors were selected.	
Week 4	Preliminary mechanical design of the quadruped robot was started.	
Week 5	Electronic system design was studied.	
Week 6	PID-based balance and walking control methods were researched.	
Week 7	Simulation studies were performed for walking and balance control algorithms.	
Week 8	**Midterm Week**	
Week 10	Mechanical parts were revised according to feedback.	
Week 11	Embedded software development was started.	
Week 12	Communication system design was completed.	
Week 13	Autonomous and semi-autonomous operation modes researched.	
Week 14	4G modem and GPS data transmission structure was implemented.	
Week 15	Balance, motion and communication performance were evaluated.	
Week 16	Final report preparation and project documentation were completed.	

Öğrencilerin veya çalışma gruplarının haftalık toplantıları sırasında bu formu proje yöneticisine imzalatmaları ve dönem sonunda proje raporuyla birlikte teslim etmek üzere saklamaları gerekmektedir.

ANNEX - 9. TUBITAK Project Closure Document

After the completion of projects such as 2209/A-B supported by TÜBİTAK, the signed version of the closure form that the advisor and students will receive from the TÜBİTAK project page should be added only to the Graduation Project as a final appendix.